

# Scenario Energetico

## Macchine e Sistemi Energetici

Laurea in Ingegneria Informatica

A.A. 2025 – 26

# Italian Energy Scenario

## Critical issues:

- Dependence on foreign supply of most primary energy sources;
- Peculiar nature of the energy production;
- Inadequacy of infrastructures
- Institutions weakness
- Deficiency in scientific and technological research

No nuclear energy due to the 1987 referendum choice

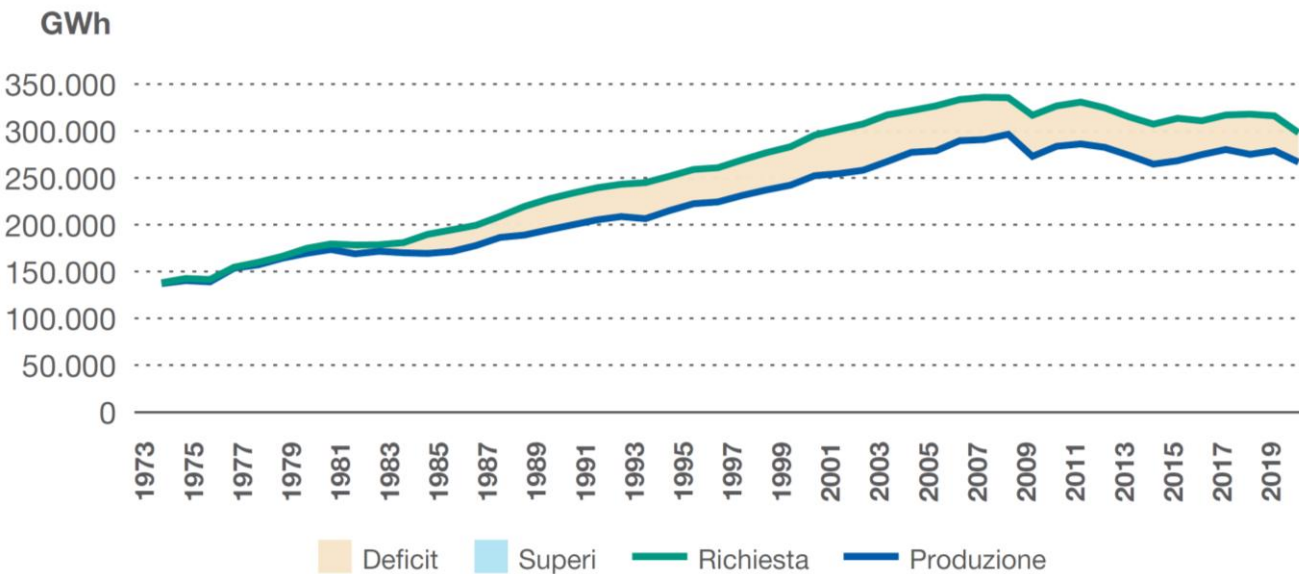
# Italian Energy Scenario

Energia richiesta in Italia nel 2020

GWh 301.180,4

Deficit (-) Superi (+) della produzione  
rispetto alla richiesta

GWh -32.200,4 (-10,7%)



Consumi anno 2020: complessivi 283.814,5 GWh; per abitante 4.777 kWh

## Terna

### Rete Elettrica Nazionale Società per Azioni

Terna is the majority owner of the Italian high voltage and very high voltage electricity National Transmission Grid. It operates in a natural monopoly and secures the transmission and dispatching of electricity throughout the country.

# Italian Energy Production and Demand

## National Electric Energy Balance in 2020

### Produzione netta

|                                                                                                                  |                                                                                   |       |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------|
| <b>Idrica: 49,0 TWh</b><br>Centrali idroelettriche                                                               |  | 16,3% |
| <b>Termica: 171,2 TWh</b><br>Centrali a olio, gas naturale, biomasse, etc.<br>(al netto di 2,7 TWh per pompaggi) |  | 56,9% |
| <b>Geotermica: 5,6 TWh</b><br>Centrali geotermoelettriche                                                        |  | 1,9%  |
| <b>Eolica e Fotovoltaica: 43,2 TWh</b><br>Centrali eoliche e fotovoltaiche                                       |  | 14,3% |

**Importazioni da estero: 32,2 TWh**



10,7%

### Consumi

|                    |                                                                                     |                    |
|--------------------|-------------------------------------------------------------------------------------|--------------------|
| <b>Agricoltura</b> |  | 6,3 TWh<br>(2%)    |
| <b>Industria</b>   |  | 125,4 TWh<br>(44%) |
| <b>Servizi</b>     |  | 85,9 TWh<br>(30%)  |
| <b>Domestico</b>   |  | 66,2 TWh<br>(23%)  |

### GSE - Gestore Servizi Elettrici

It manages the incentive mechanisms aimed at promoting the development of energy efficiency and renewable sources. It purchases the electricity fed into the grid produced by renewable energy plants to then resell it in the electricity market. It certifies the renewable origin of the electricity produced and it inspects the plants to assure a fair distribution of public resources.

**DOMANDA: 301,2 TWh**

\* Produzione netta: è la produzione lorda al netto dei servizi ausiliari

1) Include la parte biodegradabile dei rifiuti

2) Al netto dei rifiuti solidi urbani non biodegradabili, contabilizzati negli altri combustibili

3) Carbone + Lignite

4) L'energia destinata ai pompaggi viene convenzionalmente detratta dalla produzione termica non rinnovabile

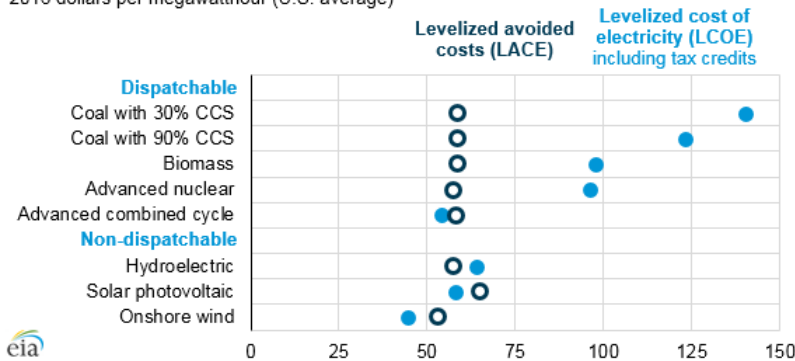
# Electricity Production Mix

## Primary Sources for Electric Energy Production

Levelized cost of energy (LCOE): common metric for cost comparisons across projects and technologies, considers a plant's expected lifetime and operation cycle and amortizes those costs over an assumed financial lifetime.

## Electric Energy Cost vs primary sources

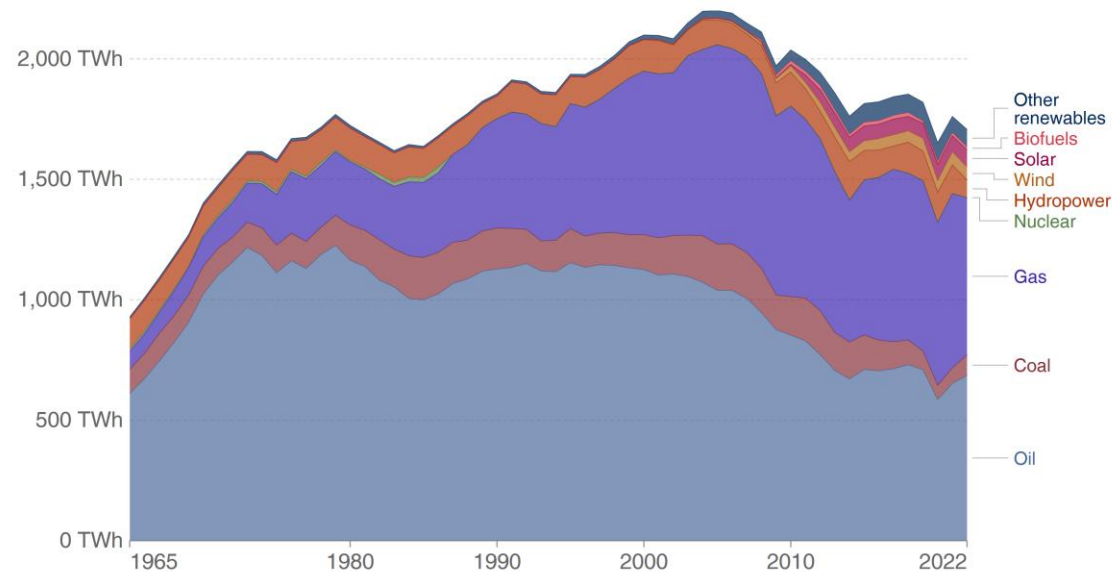
Levelized cost and levelized avoided cost of electricity projections by technology, 2022  
2016 dollars per megawatthour (U.S. average)



### Energy consumption by source, Italy

Primary energy consumption is measured in terawatt-hours (TWh). Here an inefficiency factor (the 'substitution' method) has been applied for fossil fuels, meaning the shares by each energy source give a better approximation of final energy consumption.

Our World in Data



Source: Energy Institute Statistical Review of World Energy (2023)

Note: 'Other renewables' includes geothermal, biomass and waste energy.

OurWorldInData.org/energy • CC BY

**Italy uses the most expensive primary energy sources; thus, the energy cost is high**

**Copertura della domanda totale**

**305,615.90 GWh**

↘ -3% (YoY% -2023)

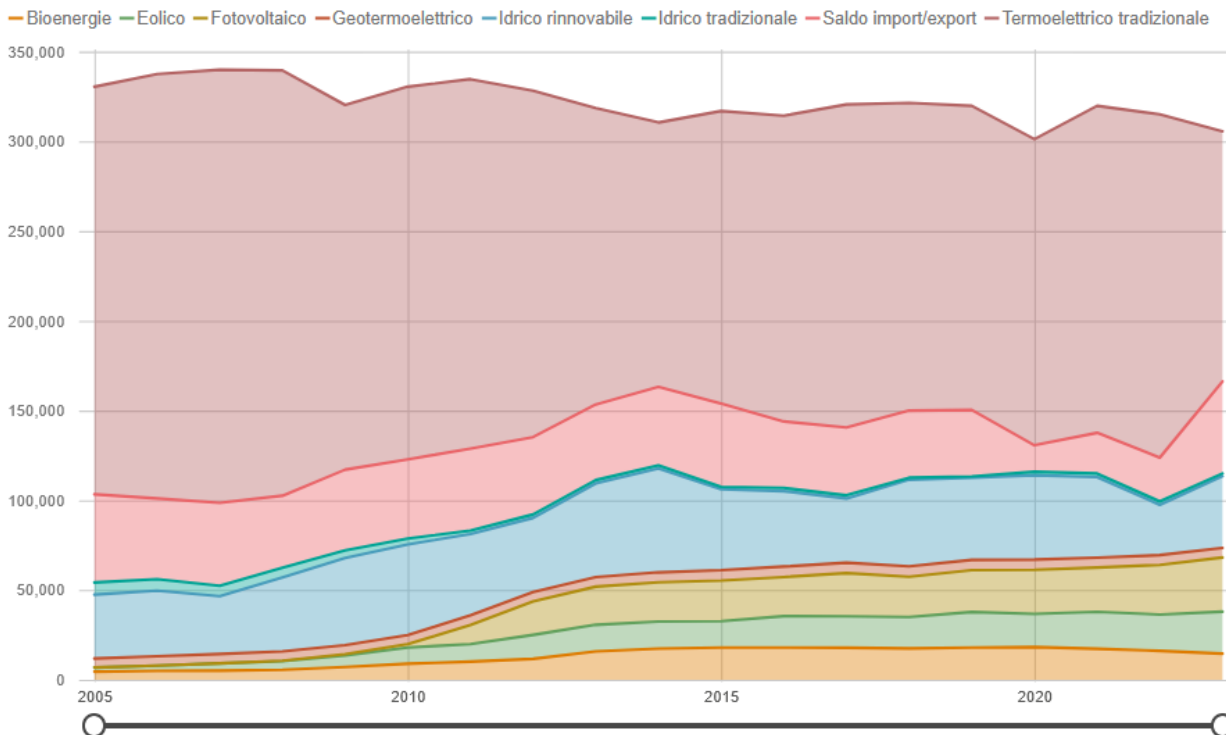
**Copertura YoY% della domanda nelle regioni**



-9% 0%

**Copertura della domanda per fonte [GWh]**

**FABBISOGNO**



**Copertura della domanda per fonte**

|                      |                  |                        |                             |                               |                  |                                 |                           |
|----------------------|------------------|------------------------|-----------------------------|-------------------------------|------------------|---------------------------------|---------------------------|
| Bioenergie<br>(4.8%) | Eolico<br>(7.6%) | Fotovoltaico<br>(9.9%) | Geotermoelettrico<br>(1.7%) | Idrico rinnovabile<br>(13.1%) | Idrico<br>(0.5%) | Saldo import/<br>export (16.8%) | Termoelettrico<br>(45.6%) |
| 14,587.14 GWh        | 23,375.10 GWh    | 30,204.92 GWh          | 5,300.00 GWh                | 39,963.27 GWh                 | 1,515.73 GWh     | 51,251.60 GWh                   | 139,418.14 GWh            |
| ↘ -9.4%              | ↗ 15.6%          | ↗ 9.2%                 | ↘ -2.8%                     | ↗ 42.6%                       | ↘ -18.6%         | ↗ 110.3%                        | ↘ -27.2%                  |

**Produzione di energia totale**

**264,708.09 GWh**

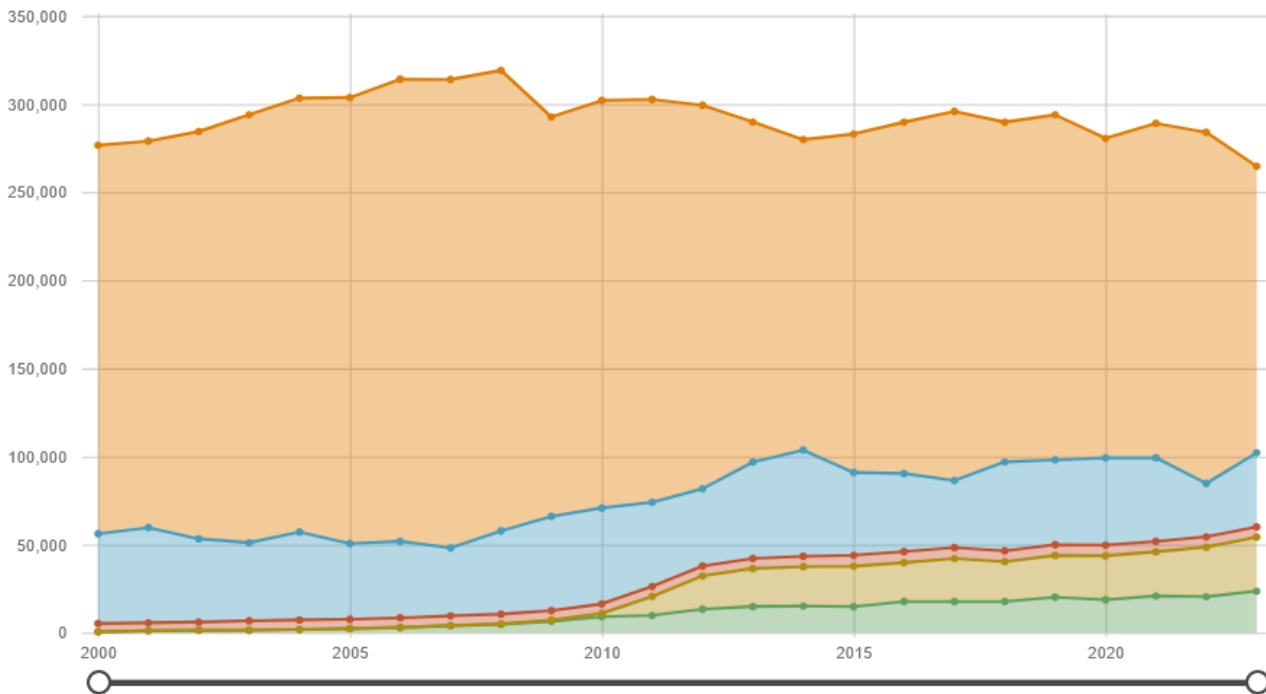
↘ -6.8% (YoY% - 2023)

**Produzione YoY% nelle regioni**



**Produzione per fonte [GWh]**

● Eolico ● Fotovoltaico ● Geotermoelettrico ● Idrico ● Termoelettrico



**Produzione per fonte**

|                      |                           |                               |                      |                             |
|----------------------|---------------------------|-------------------------------|----------------------|-----------------------------|
| ● Eolico<br>(8.9%)   | ● Fotovoltaico<br>(11.6%) | ● Geotermoelettrico<br>(2.2%) | ● Idrico<br>(15.9%)  | ● Termoelettrico<br>(61.4%) |
| <b>23,640.47 GWh</b> | <b>30,711.08 GWh</b>      | <b>5,700.55 GWh</b>           | <b>42,068.26 GWh</b> | <b>162,587.72 GWh</b>       |
| ↗ 15.4%              | ↗ 9.3%                    | ↘ -2.4%                       | ↗ 38.9%              | ↘ -18.4%                    |



**Potenza efficiente per fonte**

**129,943.89 MW**

↗ 5.4% (YoY% - 2023)

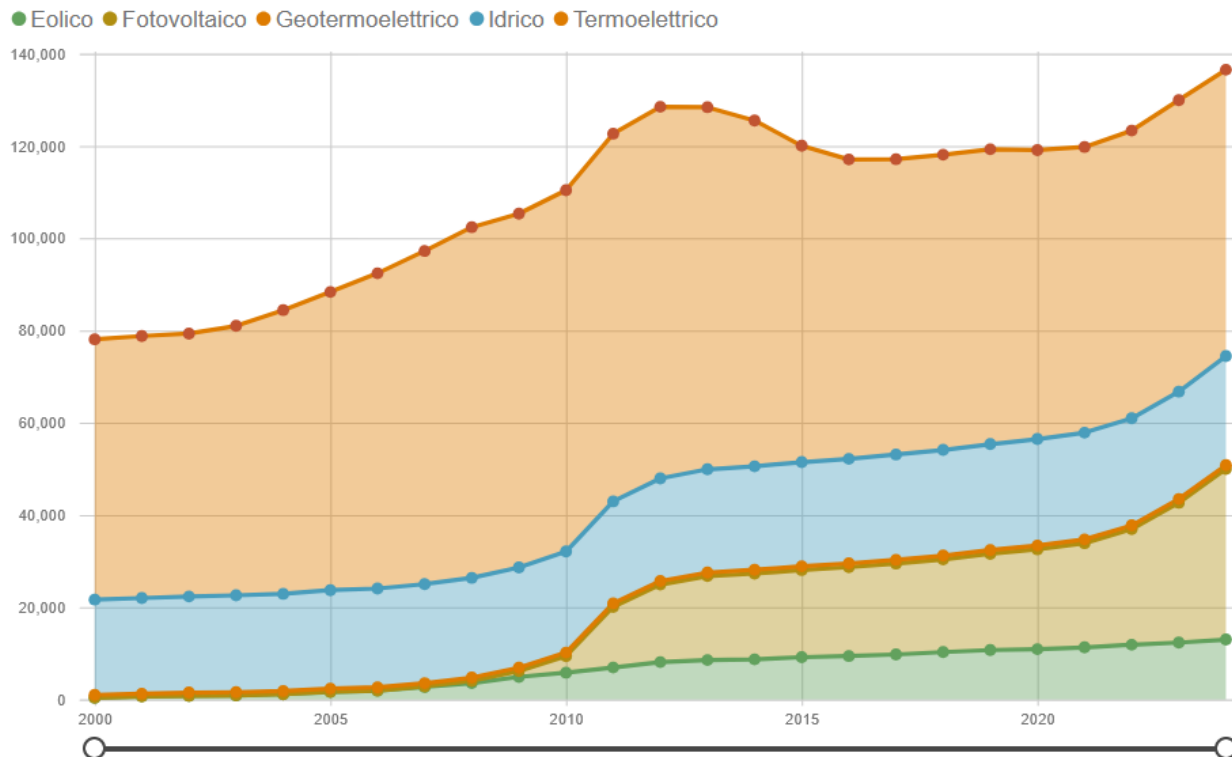
**Potenza efficiente YoY% nelle regioni**



1%  11%

**Potenza efficiente per fonte [MW]**

**CAPACITA' IMPIANTI DI GENERAZIONE**



**Potenza efficiente per fonte**

● Eolico  
(9.5%)

**12,335.54 MW**

↗ 4.1%

● Fotovoltaico  
(23.3%)

**30,319.42 MW**

↗ 21%

● Geotermoelettrico  
(0.6%)

**817.09 MW**

→ 0%

● Idrico  
(17.9%)

**23,260.47 MW**

↗ 0.3%

● Termoelettrico  
(48.6%)

**63,211.37 MW**

↗ 1.4%



**Produzione totale**

**116,578.65 GWh**

↑ 16.1% (YoY% - 2023)

**Produzione da fonti rinnovabili YoY% nelle regioni**

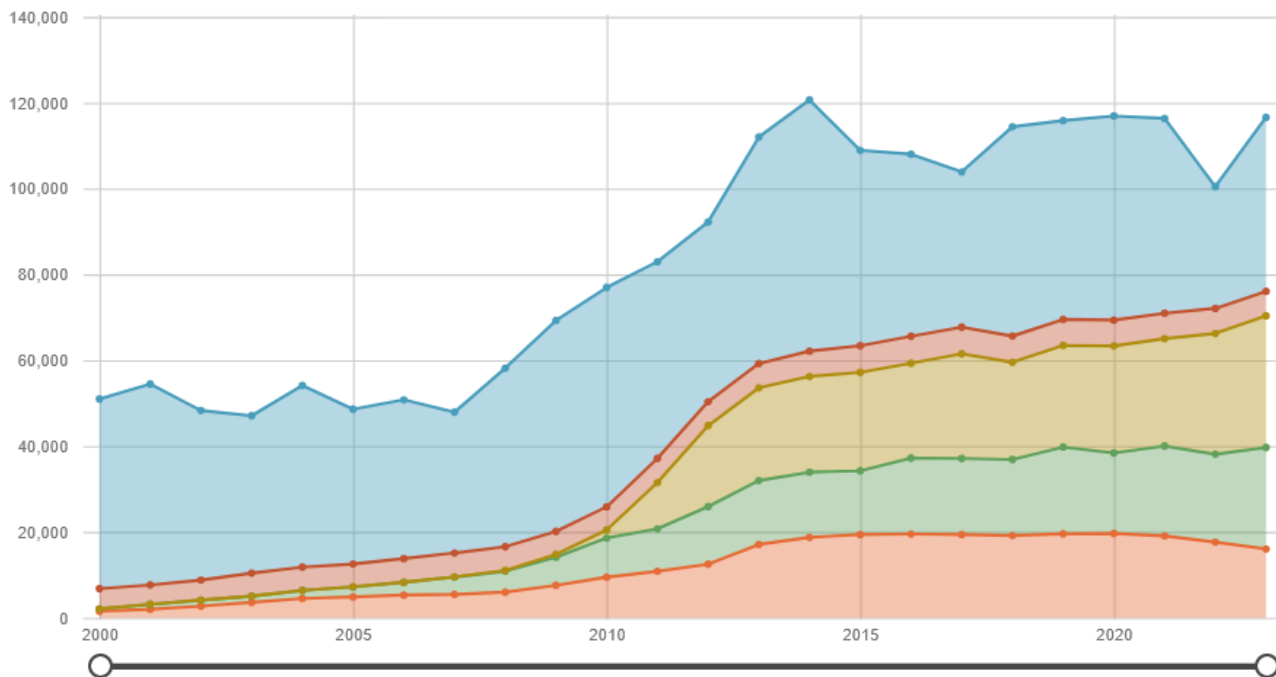


-3%

46%

**Produzione per fonte rinnovabile [GWh]**

● Bioenergie ● Eolico ● Fotovoltaico ● Geotermoelettrico ● Idrico



**Produzione per fonte rinnovabile**

● Bioenergie  
(13.7%)

**16,017.64 GWh**

↓ -9.1%

● Eolico  
(20.3%)

**23,640.47 GWh**

↑ 15.4%

● Fotovoltaico  
(26.3%)

**30,711.08 GWh**

↑ 9.3%

● Geotermoelettrico  
(4.9%)

**5,692.19 GWh**

↓ -2.5%

● Idrico  
(34.8%)

**40,517.26 GWh**

↑ 42.7%

**Potenza efficiente totale**

**66,824.98 MW**

↗ 9.5% (YoY% - 2023)

**Potenza efficiente YoY% nelle regioni**

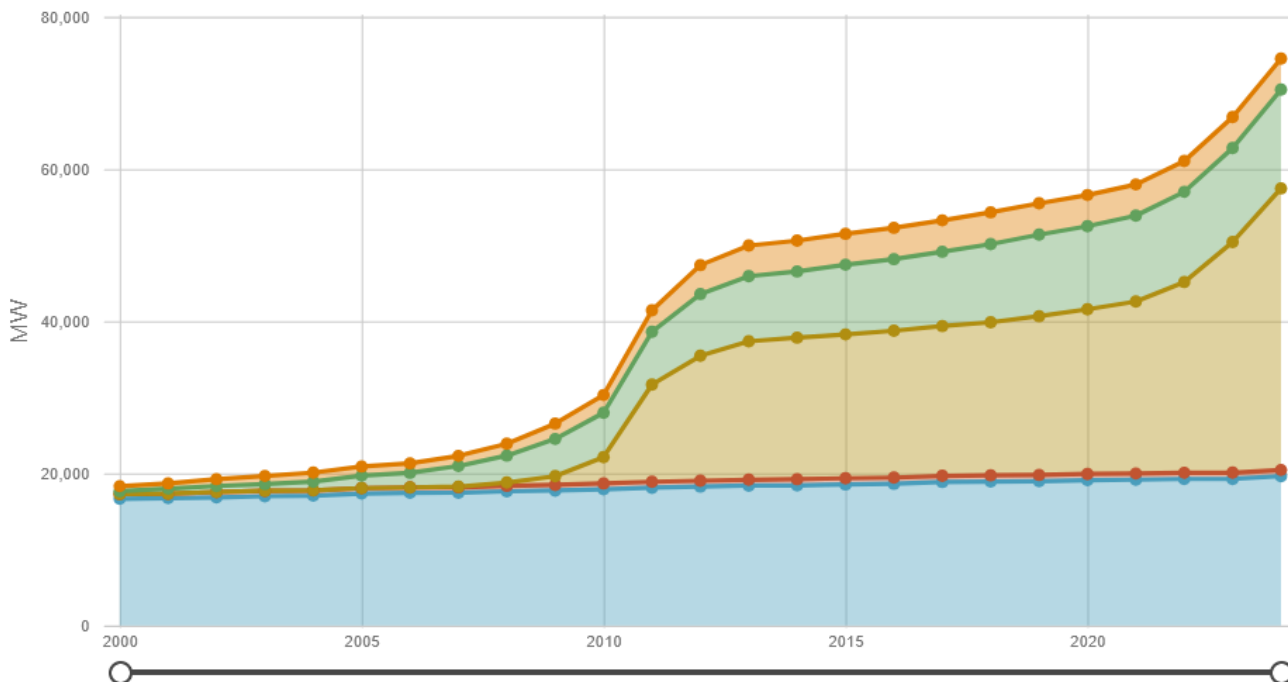


1% 17%

**Potenza efficiente nazionale per fonte rinnovabile [MW]**

**CAPACITA' FONTI RINNOVABILI**

● Idrico ● Geotermoelettrico ● Fotovoltaico ● Eolico ● Bioenergie



**Potenza efficiente per fonte rinnovabile**

● Idrico  
(28.8%)

**19,274.17 MW**

↗ 0.1%

● Eolico  
(18.5%)

**12,335.54 MW**

↗ 4.1%

● Geotermoelettrico  
(1.2%)

**817.09 MW**

→ 0%

● Fotovoltaico  
(45.4%)

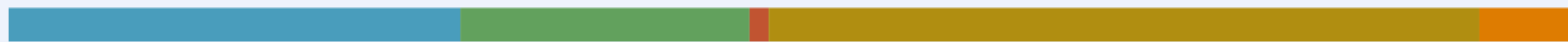
**30,319.42 MW**

↗ 21%

● Bioenergie  
(6.1%)

**4,078.76 MW**

↗ 0.8%



**Produzione lorda totale**

**162,587.72 GWh**

↘ -18.4% (YoY% - 2023)

**Produzione lorda YoY% nelle regioni**



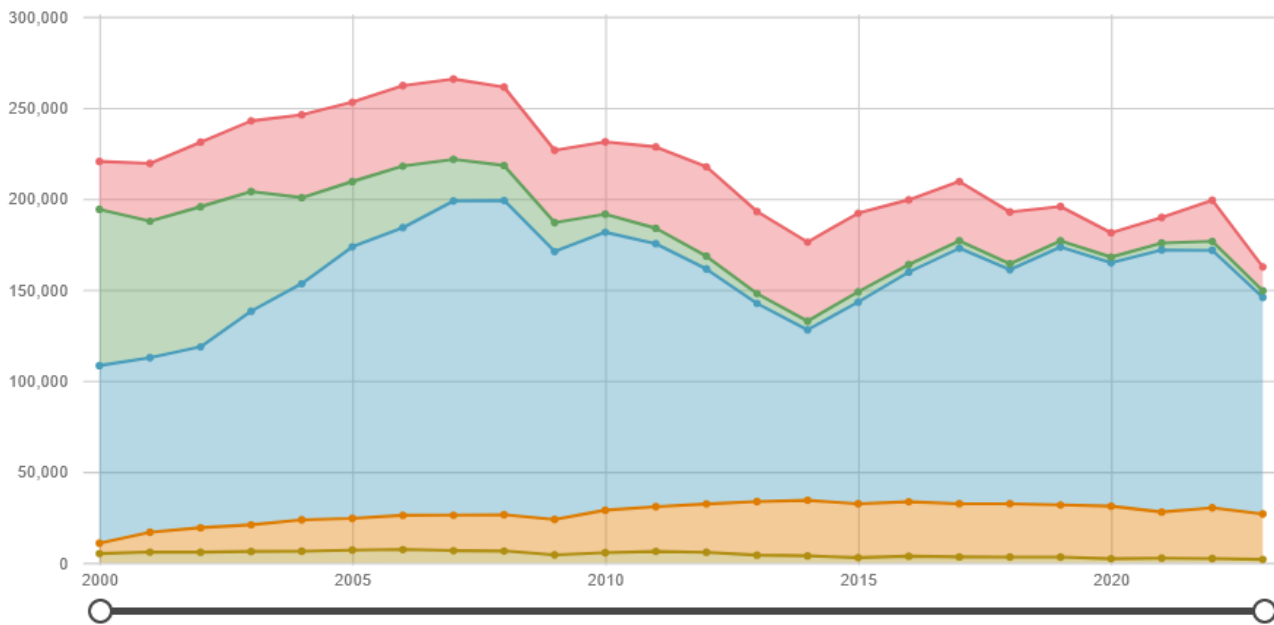
-61%

22%

**PRODUZIONE ENERGIA PER COMBUSTIBILE**

**Produzione lorda per combustibile [GWh]**

● Altri ● Bioenergie ● Gas Naturale ● Petroliferi ● Solidi



**Produzione lorda per combustibile**

|                     |                         |                           |                         |                      |
|---------------------|-------------------------|---------------------------|-------------------------|----------------------|
| ● Altri<br>(1.1%)   | ● Bioenergie<br>(15.4%) | ● Gas Naturale<br>(73.2%) | ● Petroliferi<br>(2.2%) | ● Solidi<br>(8.1%)   |
| <b>1,794.85 GWh</b> | <b>24,964.08 GWh</b>    | <b>118,986.51 GWh</b>     | <b>3,622.39 GWh</b>     | <b>13,219.89 GWh</b> |
| ↘ -37.9%            | ↘ -5.6%                 | ↘ -15.8%                  | ↘ -30%                  | ↘ -42.6%             |

**67.22 MtCO<sub>2</sub>**

↘ -18.4% (YoY% - 2023)

**Emissione CO<sub>2</sub> (mln di tonnellate) YoY%  
nelle regioni**



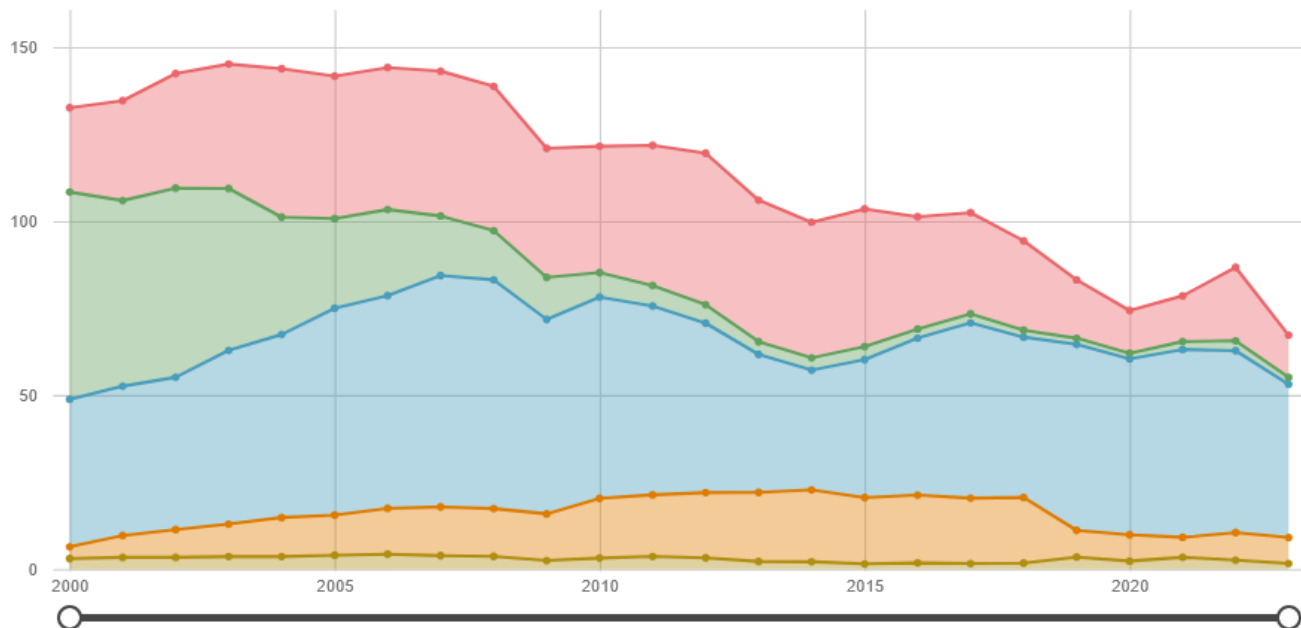
62%

31%

## EMISSIONI CO<sub>2</sub> PER COMBUSTIBILE

Emissione per combustibile [mln di tonnellate]

● Altri ● Bioenergie ● Gas Naturale ● Petroliferi ● Solidi



**Emissione per combustibile**

● Altri  
(2.4%)

**1.59 MtCO<sub>2</sub>**

↘ -37.9%

● Bioenergie  
(11.1%)

**7.48 MtCO<sub>2</sub>**

↘ -5.6%

● Gas Naturale  
(65.5%)

**44.03 MtCO<sub>2</sub>**

↘ -15.8%

● Petroliferi  
(3.0%)

**2.01 MtCO<sub>2</sub>**

↘ -22.5%

● Solidi  
(18.0%)

**12.10 MtCO<sub>2</sub>**

↘ -22.5%

## Total Load

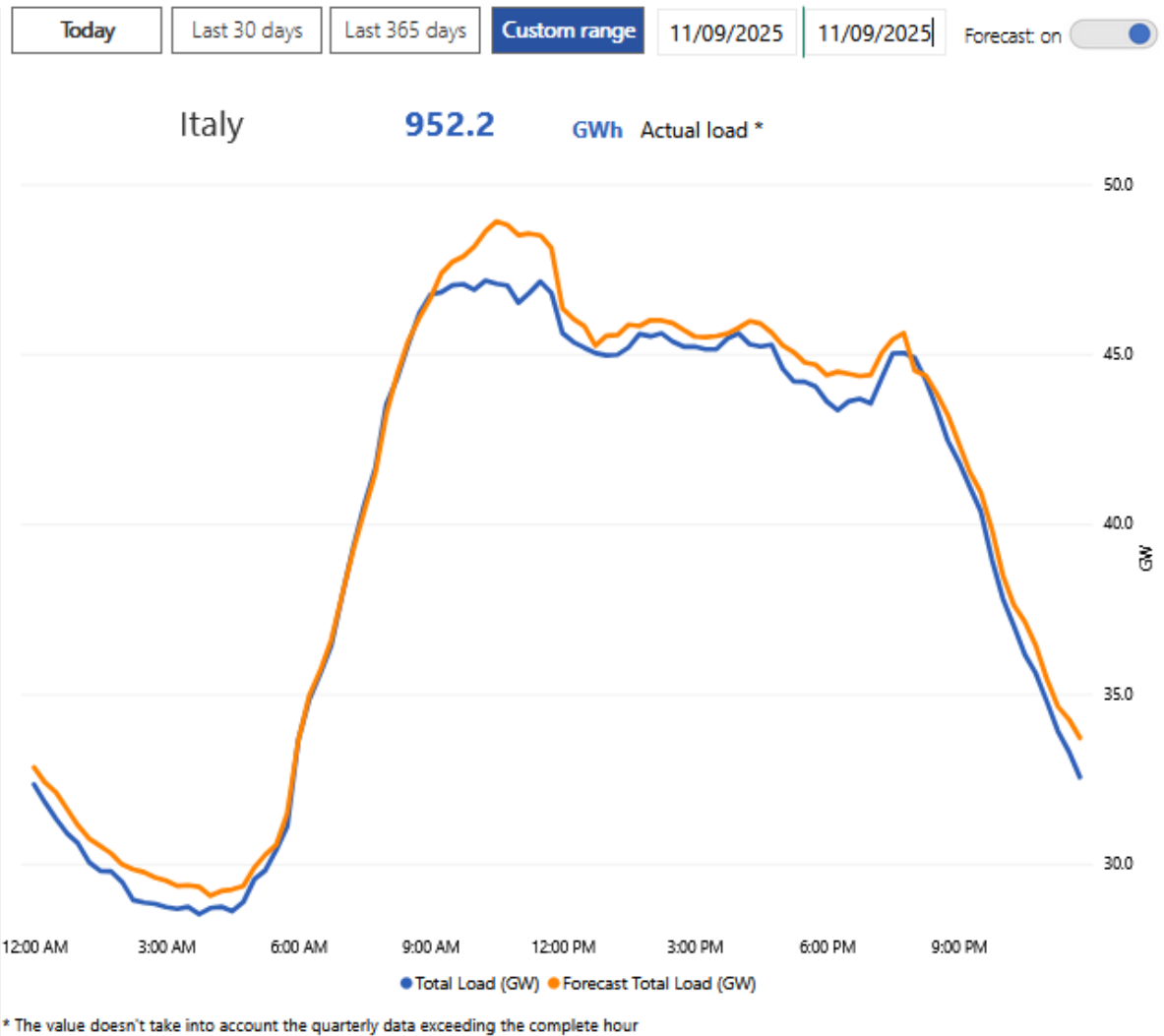
From: **11/09/2025** To: **11/09/2025**

Last update: 12/09/2025 17:00



Actual / Forecast load  
per bidding-zone [GWh]

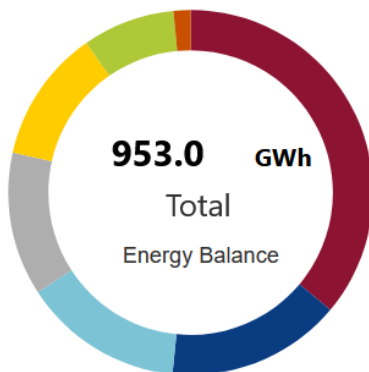
|              |       |
|--------------|-------|
| North        | 534.8 |
| Centre-South | 159.0 |
| Centre-North | 81.0  |
| South        | 74.0  |
| Sicily       | 62.5  |
| Sardinia     | 22.9  |
| Calabria     | 18.1  |



## Energy Balance

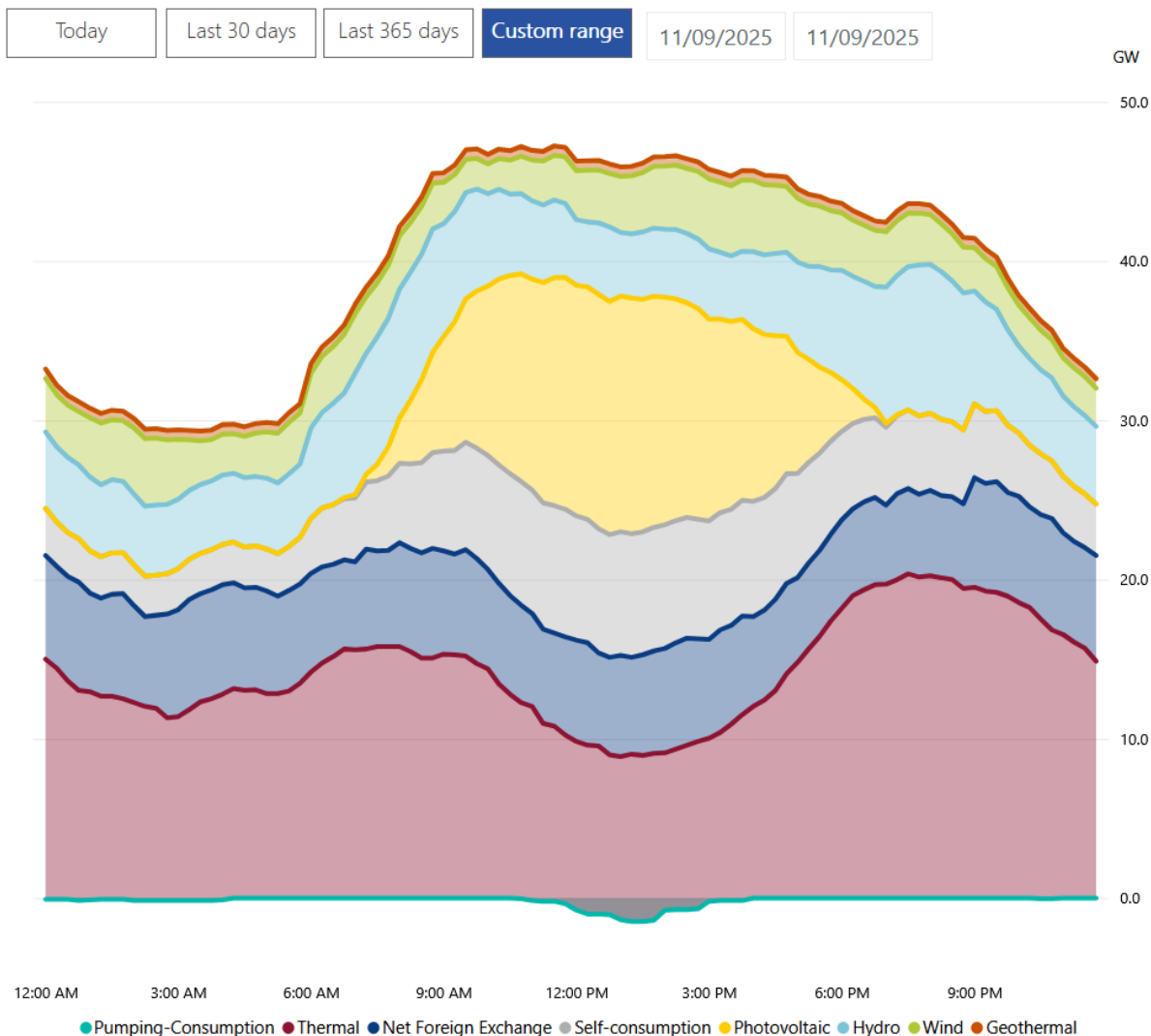
From: **11/09/2025** To: **11/09/2025**

Last update: 12/09/2025 15:45



Energy Balance  
per energy sources [GWh]

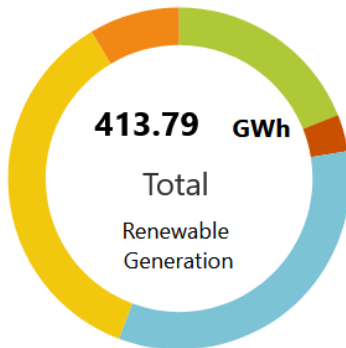
|                      |       |  |
|----------------------|-------|--|
| Thermal              | 345.7 |  |
| Net Foreign Exchange | 148.0 |  |
| Hydro                | 136.1 |  |
| Self-consumption     | 121.7 |  |
| Photovoltaic         | 112.7 |  |
| Wind                 | 78.4  |  |
| Geothermal           | 14.4  |  |
| Pumping-Consumption  | -4.1  |  |



## Actual Renewable Generation

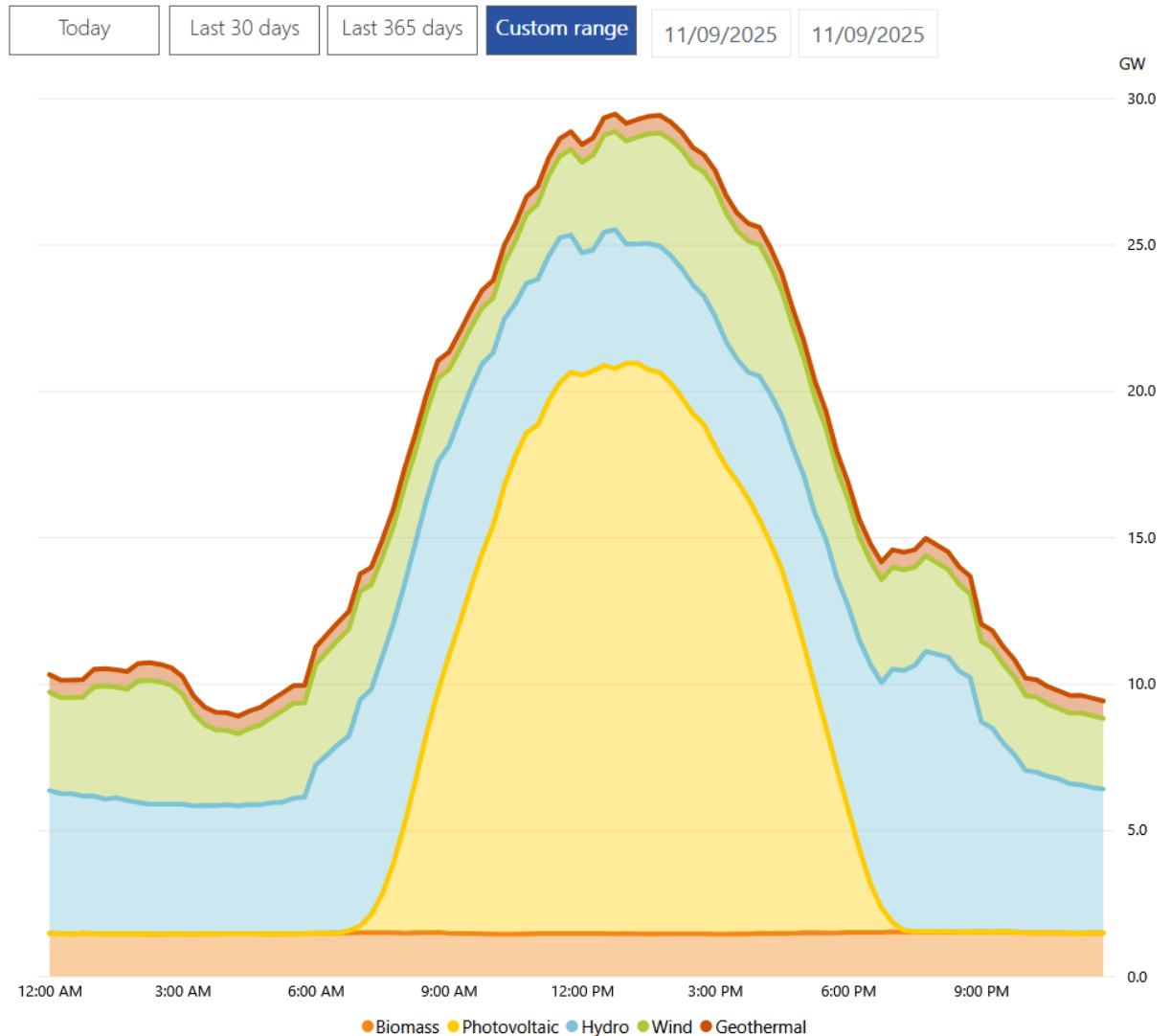
From: **11/09/2025** To: **11/09/2025**

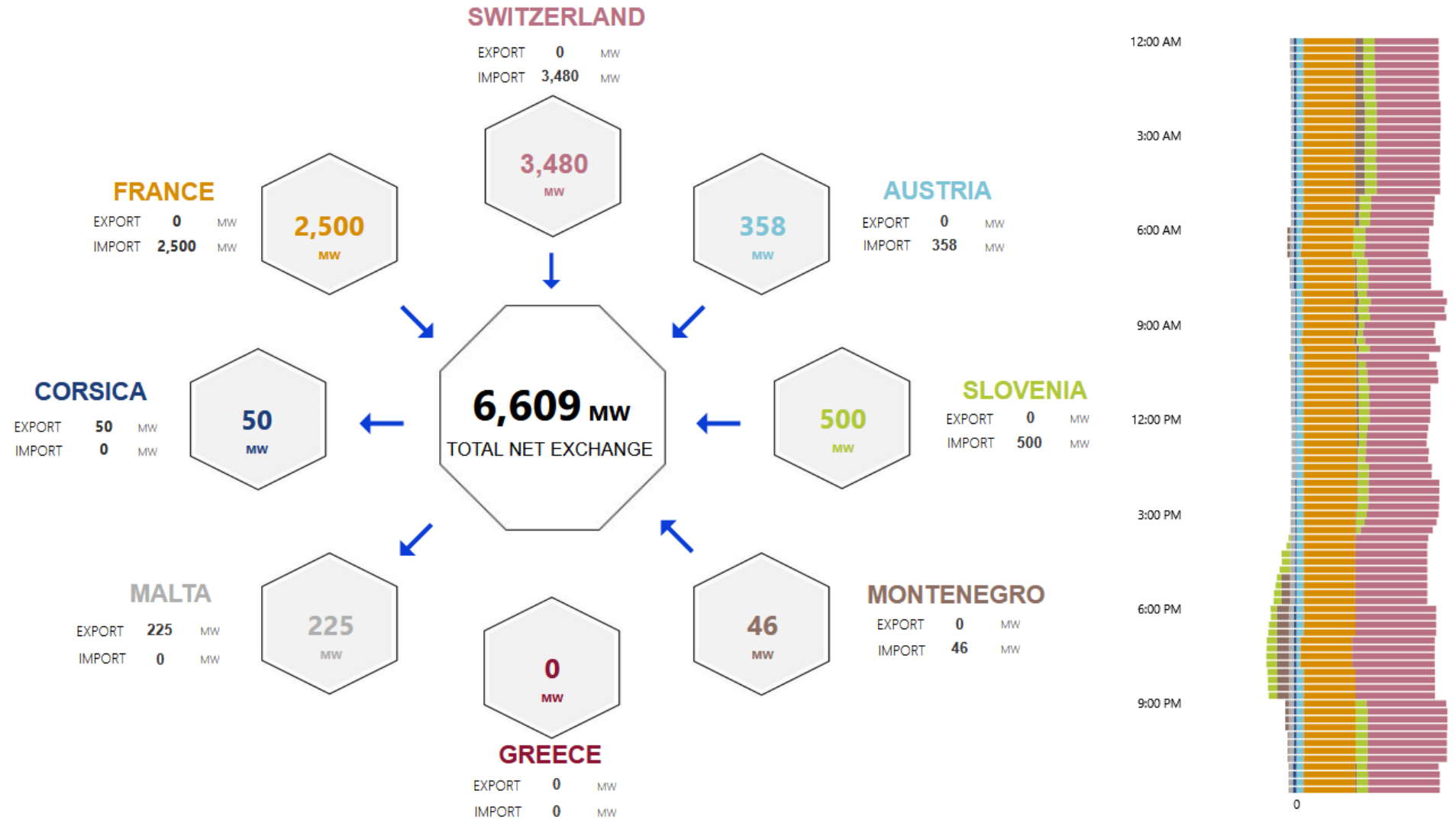
Last update: 11/09/2025 23:45



### Actual Renewable Generation per energy source [GWh]

|              |         |  |
|--------------|---------|--|
| Photovoltaic | 147.884 |  |
| Hydro        | 137.772 |  |
| Wind         | 78.387  |  |
| Biomass      | 35.347  |  |
| Geothermal   | 14.4    |  |





# Possible solutions?

## Production



### Cost reduction

- Coal
- Nuclear?
- Renewables

### Polygeneration:

- Wider energy mix
- Supply flexibility

## Demand



### Lower and better consumption

- Energy efficiency
- Good behaviour

# Possible solutions?

## Production



### Cost reduction

- Coal
- Nuclear?
- Renewables

### Polygeneration:

- Wider energy mix
- Supply flexibility

## Demand



### Lower and better consumption

- Energy efficiency
- Good behaviour

# Possible solutions?

## European Community objectives:

**The 2020 package is a set of binding legislation to ensure the EU meets its climate and energy targets for the year 2020.**

**The package sets three key targets:**



The targets were set by EU leaders in 2007 and enacted in legislation in 2009. They are also headline targets of the Europe 2020 strategy for smart, sustainable and inclusive growth.

The EU is taking action in several areas to meet the targets.

**20% cut in  
greenhouse  
gas emissions**

**20% of EU  
energy from  
renewables**

**20%  
improvement  
in energy  
efficiency**

*(With respect to 1990)*

# If we stop?

## European Community 2030 climate & energy framework:

- At least 40% cuts in greenhouse gas emissions (from 1990 levels)
- At least 32% share for renewable energy
- At least 32.5% improvement in energy efficiency

The framework was adopted by EU leaders in October 2014. It builds on the 2020 climate and energy package.

It is also in line with the longer term perspective set out in the Roadmap for moving to a competitive low carbon economy in 2050, the Energy Roadmap 2050 and the Transport White Paper.

## ... and next?

**taking the next step towards the goal of reducing greenhouse gas emissions by 80-95% below 1990 level by 2050**

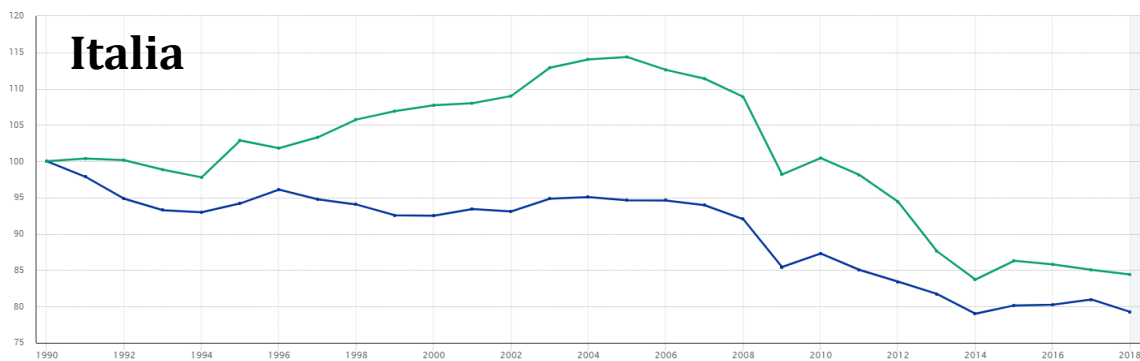
The EU aims to be climate-neutral by 2050 – an economy with net-zero greenhouse gas emissions. This objective is at the heart of the European Green Deal and in line with the EU's commitment to global climate action under the Paris Agreement.

# If we stop?

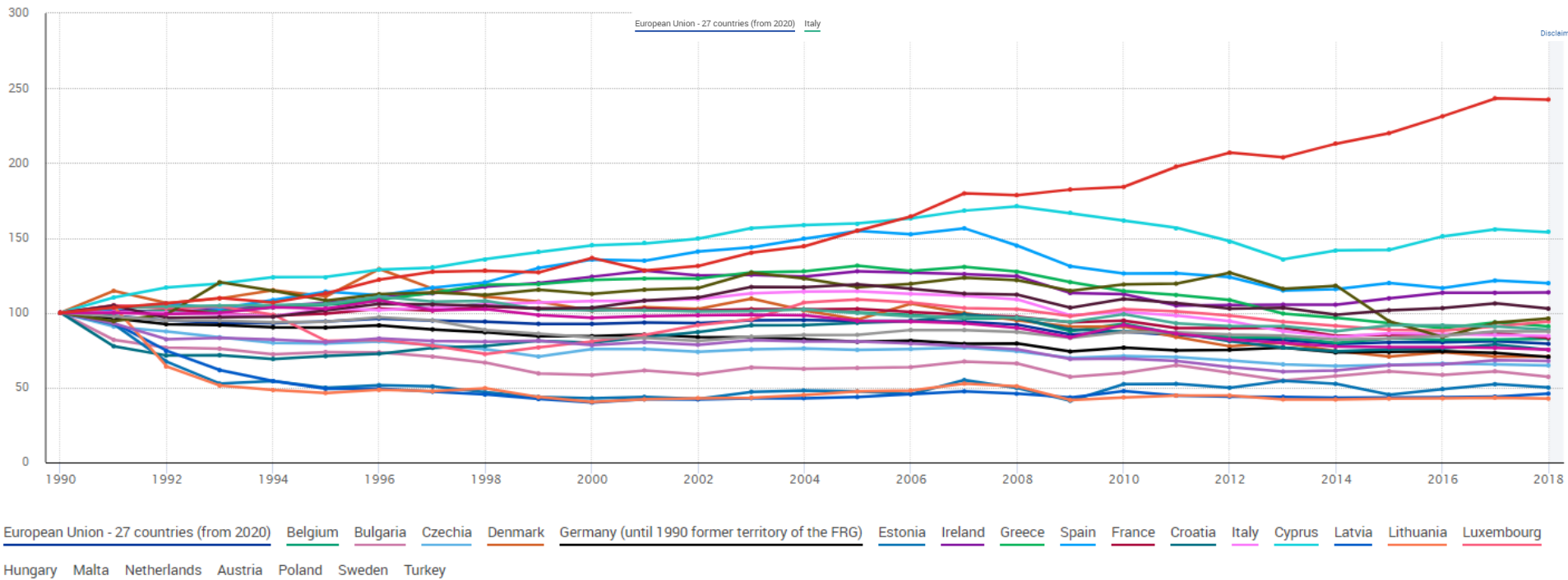
**Greenhouse gas emissions, base year 1990**  
**Index (1990 = 100)**

European Union - 27 countries (from 2020)  
2018: 79.26 / ● Target (TARGET): - (Δ: -)  
Annual, Greenhouse gas emissions

Italy  
2018: 84.41 / ● Target (TARGET): - (Δ: -)  
Annual, Greenhouse gas emissions



European Union - 27 countries (from 2020) Italy



Disclaimer

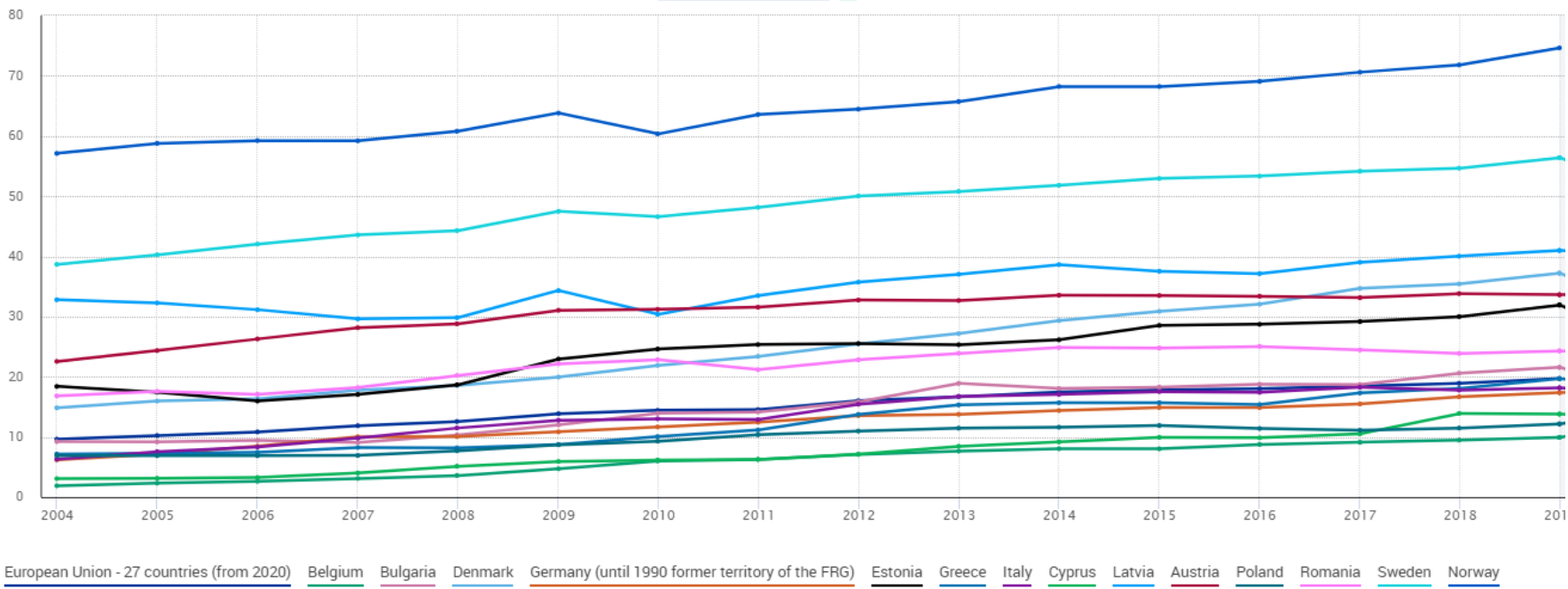
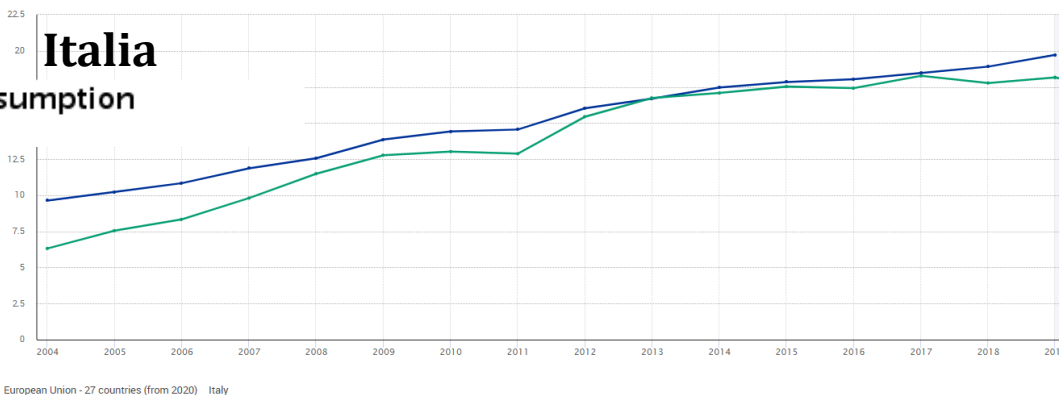
# If we stop?

## Share of renewable energy in gross final energy consumption

%

European Union - 27 countries (from 2020)  
2019: 19.727 / Target (TARGET): - (Δ: -)  
Annual, Percentage, Share of renewables

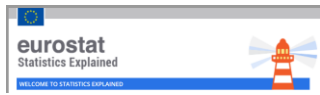
Italy  
2019: 18.161 / Target (TARGET): 17 (Δ: -1.161)  
Annual, Percentage, Share of renewables



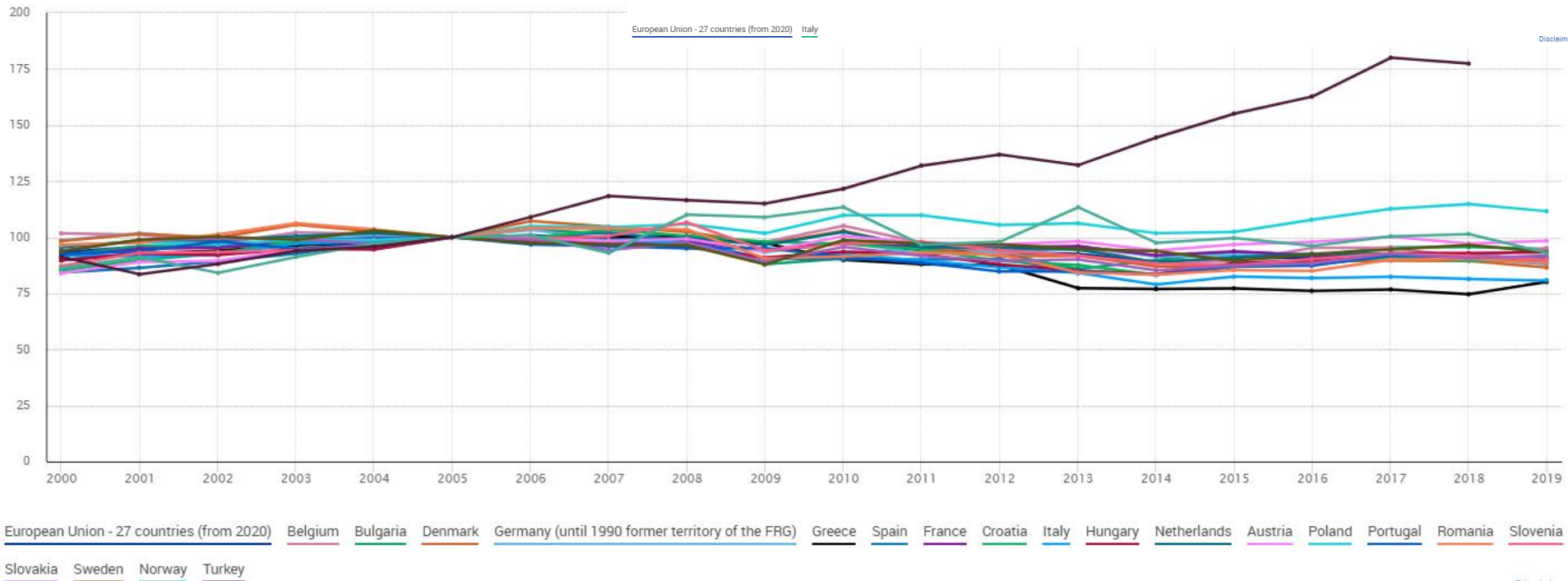
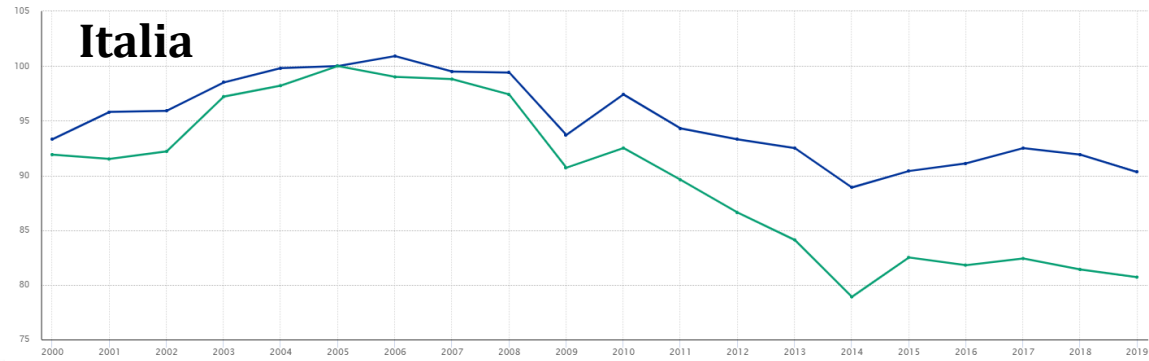
# If we stop?

**Primary energy consumption**  
**Index, 2005=100**

European Union - 27 countries (from 2020)  
2019: 90.3  
Annual, Index, 2005=100



Italy  
2019: 80.7  
Annual, Index, 2005=100



## ***IEA***

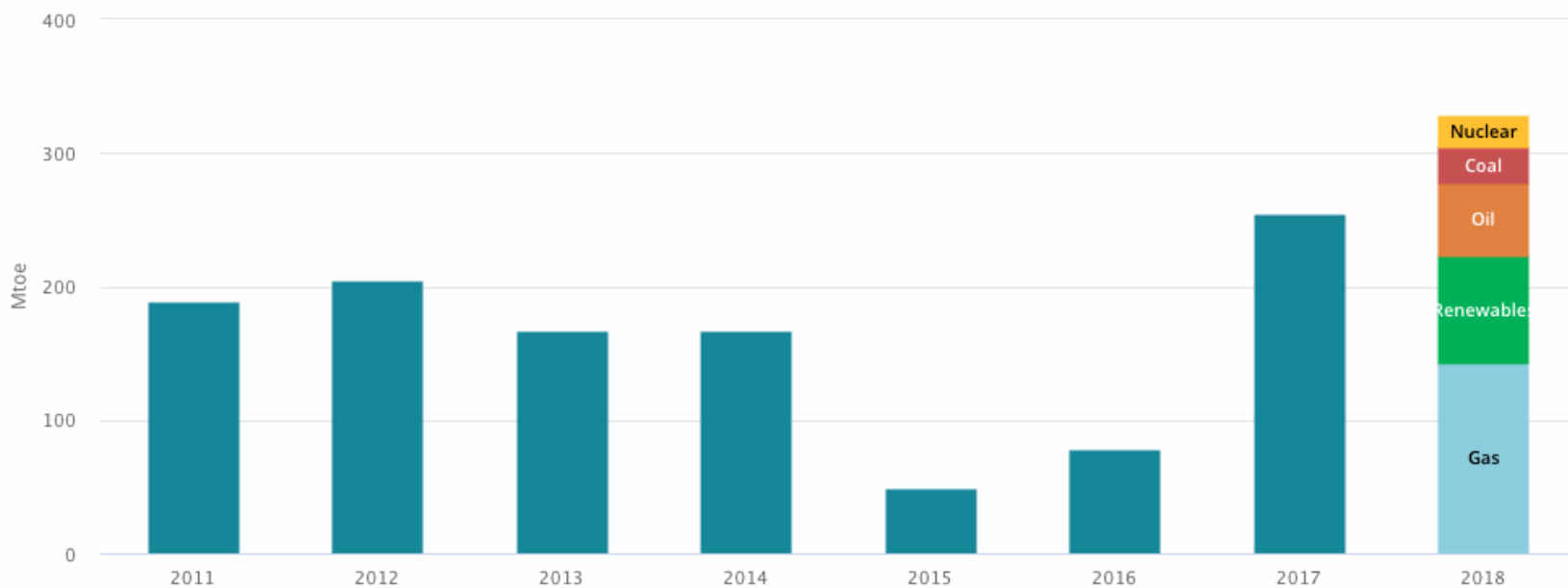
- ***GLOBAL ENERGY & CO2 STATUS REPORT 2019***
- ***CO2 EMISSIONS IN 2023***

## IEA - Global Energy & CO2 Status Report (2019)

### ► Global trends

Energy consumption worldwide grew by 2.3% in 2018

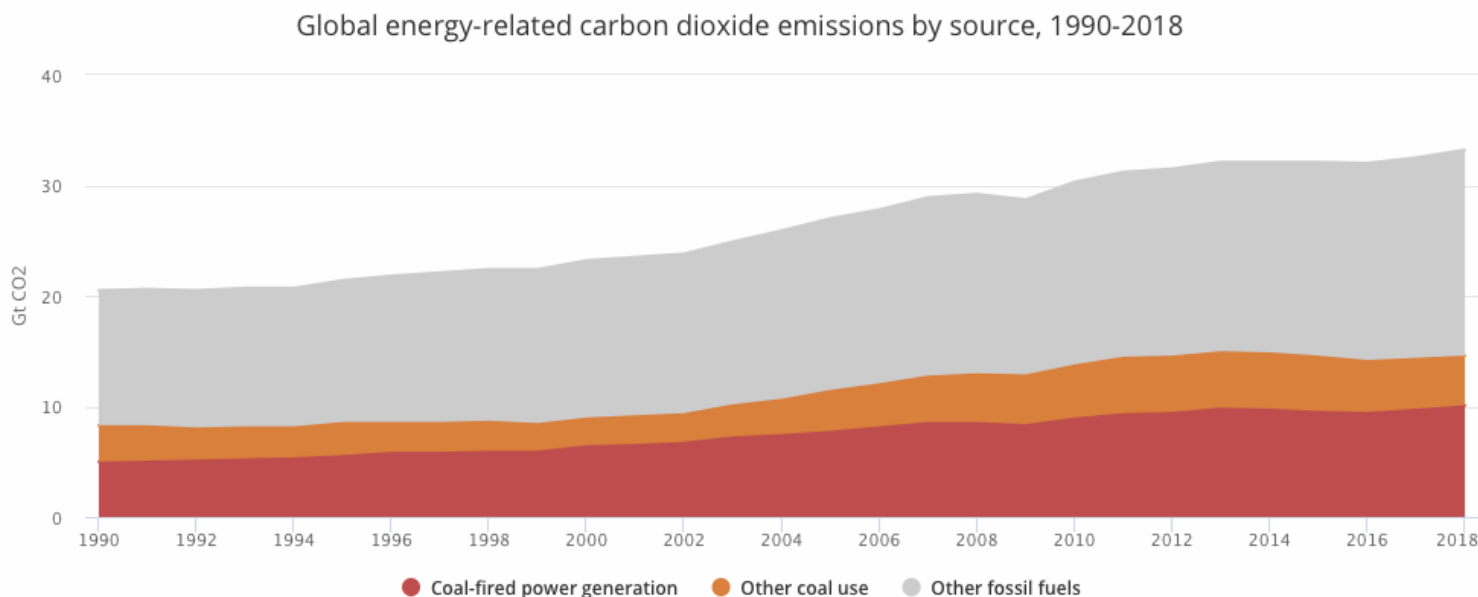
Annual change in global primary energy demand, 2011-18



- **Global energy consumption** in 2018 **increased 2.3% in 2018**
- The growth is driven by a robust global economy (+3.7%)

## IEA - Global Energy & CO2 Status Report (2019)

### ► CO<sub>2</sub> Emissions



- Global energy-related CO<sub>2</sub> emissions rose, due to higher energy demand
- Emissions from all fossil fuels increased
- Power sector: 2/3 of emissions growth
- China, India, and the US: 85% of the net increase in emissions
- Emissions declined for Germany, Japan, Mexico, France and the UK

## IEA - Global Energy & CO2 Status Report (2019)

### ► CO<sub>2</sub> Emissions

#### COAL

For the first time, **IEA assessed the impact of fossil fuel use on global temperature increases:**

- CO<sub>2</sub> from **coal combustion** was responsible for **> 0.3°C of the 1°C increase** in global average annual surface temperatures above pre-industrial levels.
- Coal is the largest source of global temperature increase.

⇒ coal-fired power plants were the single largest contributor to the growth in emissions in 2018

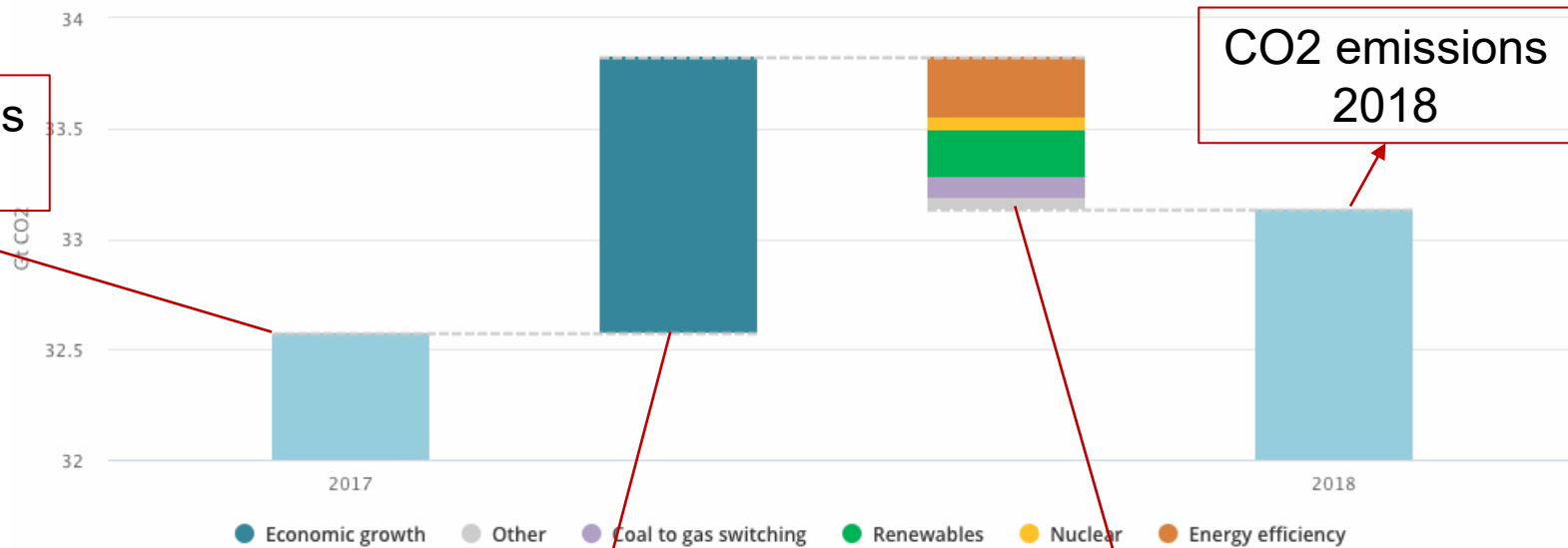
⇒ coal-fired electricity generation accounted for 30% of global CO<sub>2</sub> emissions

- !! Despite growth in coal use, **fuel switching between coal and gas** accelerated in 2018
- **Coal-to-gas switching avoided almost 60 Mt** of coal demand
- Without coal-to-gas switch, the increase in emissions would have been more **than 15% greater**

## IEA - Global Energy & CO2 Status Report (2019)

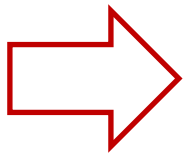
### ► CO<sub>2</sub> Emissions

Change in global energy related CO<sub>2</sub> emissions and avoided emissions, 2017-18



Economic growth

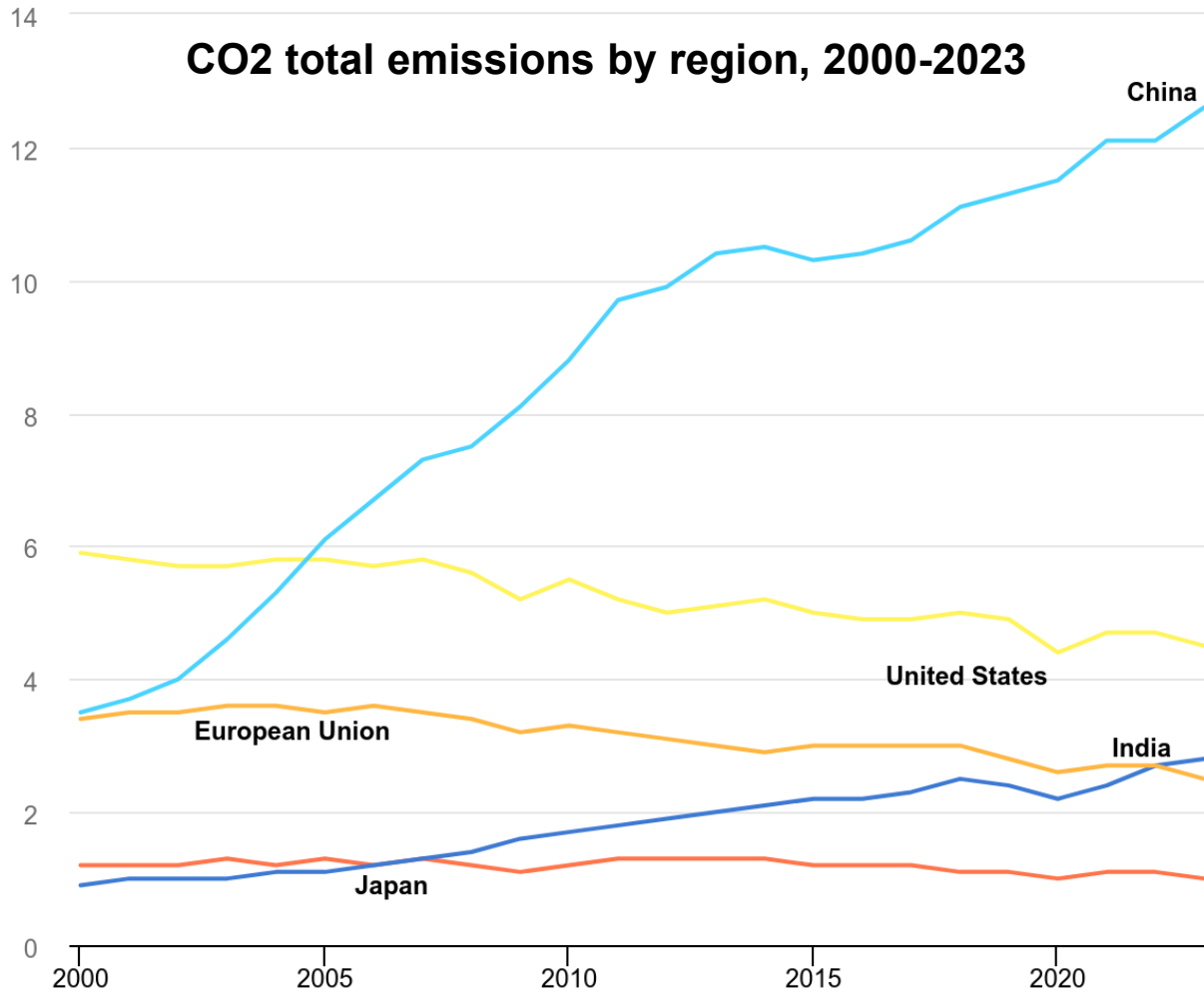
Avoided emissions



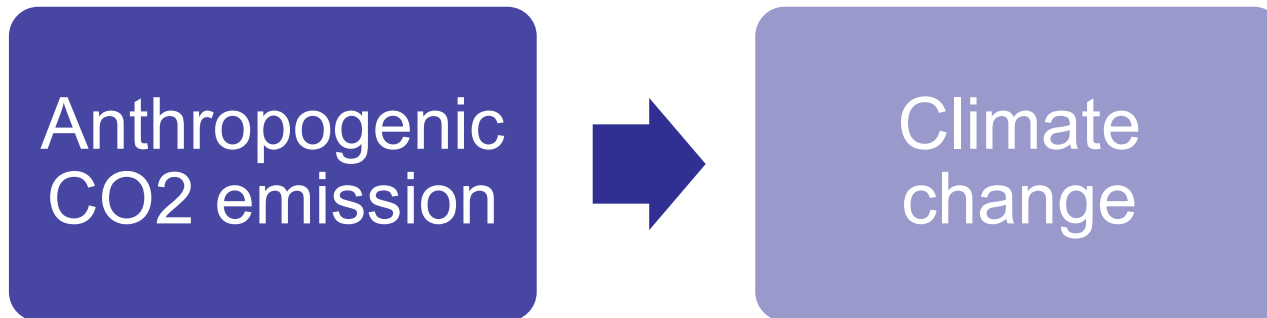
**The energy efficiency increase was not fast enough to keep pace with the rapid growth in electricity demand (+ 1 000 TWh!)**

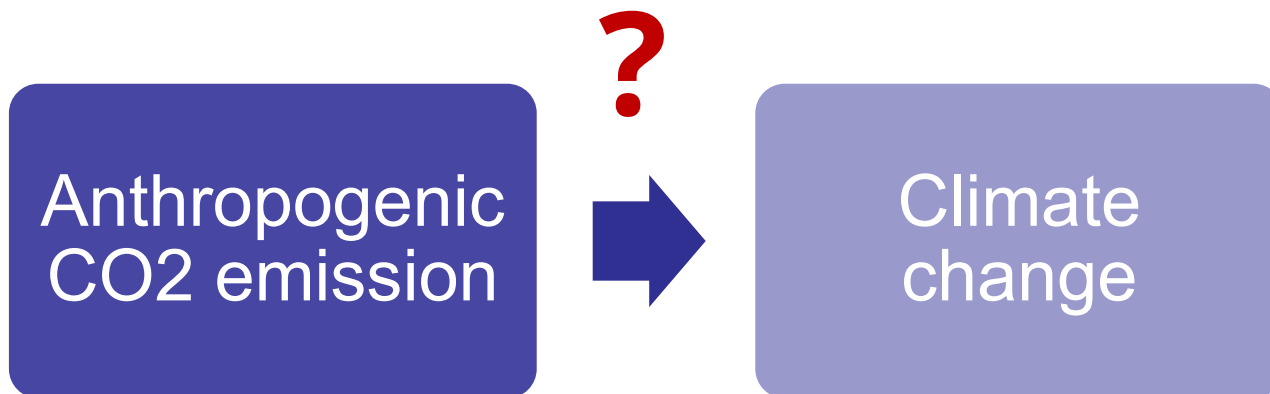
Gt CO<sub>2</sub>

## CO<sub>2</sub> total emissions by region, 2000-2023

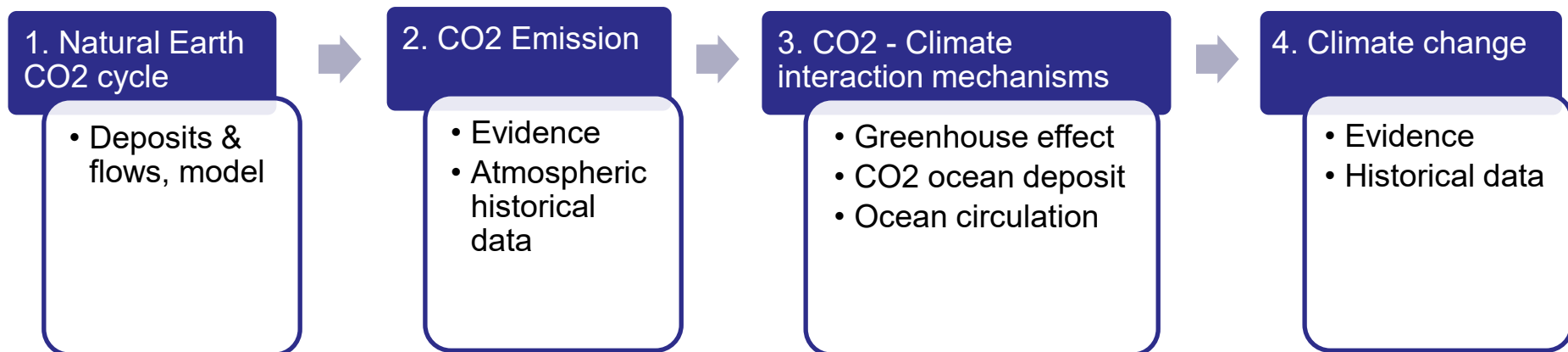


# ***CO<sub>2</sub> EMISSION SURVEY***



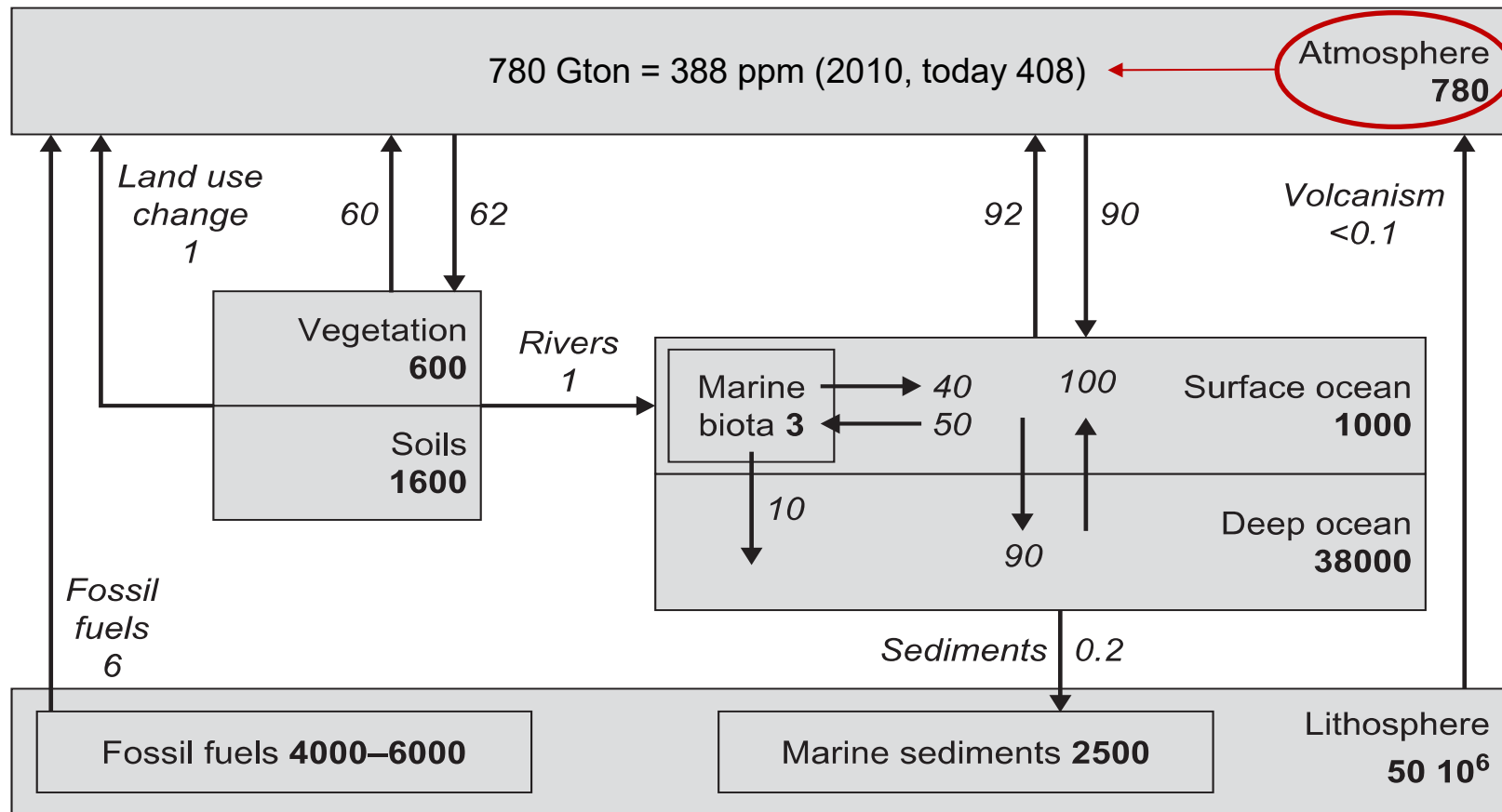


**What do we need to demonstrate it?**



1. Natural Earth  
CO<sub>2</sub> cycle

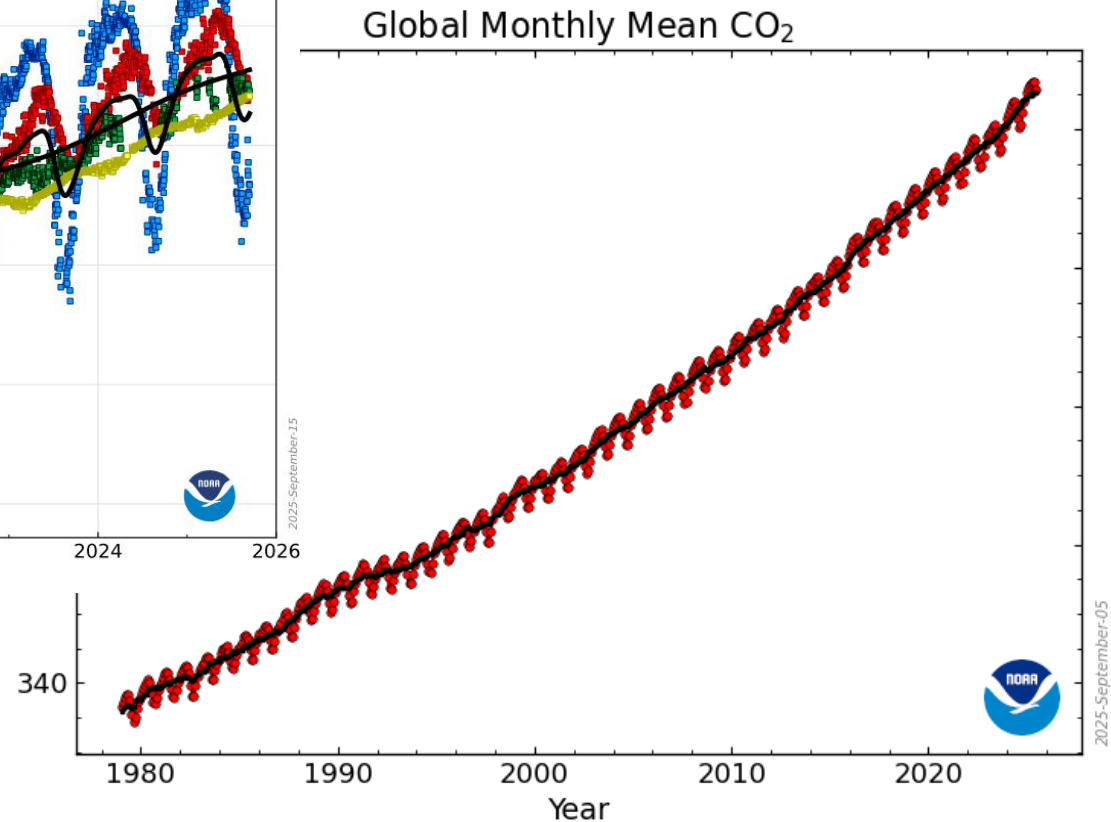
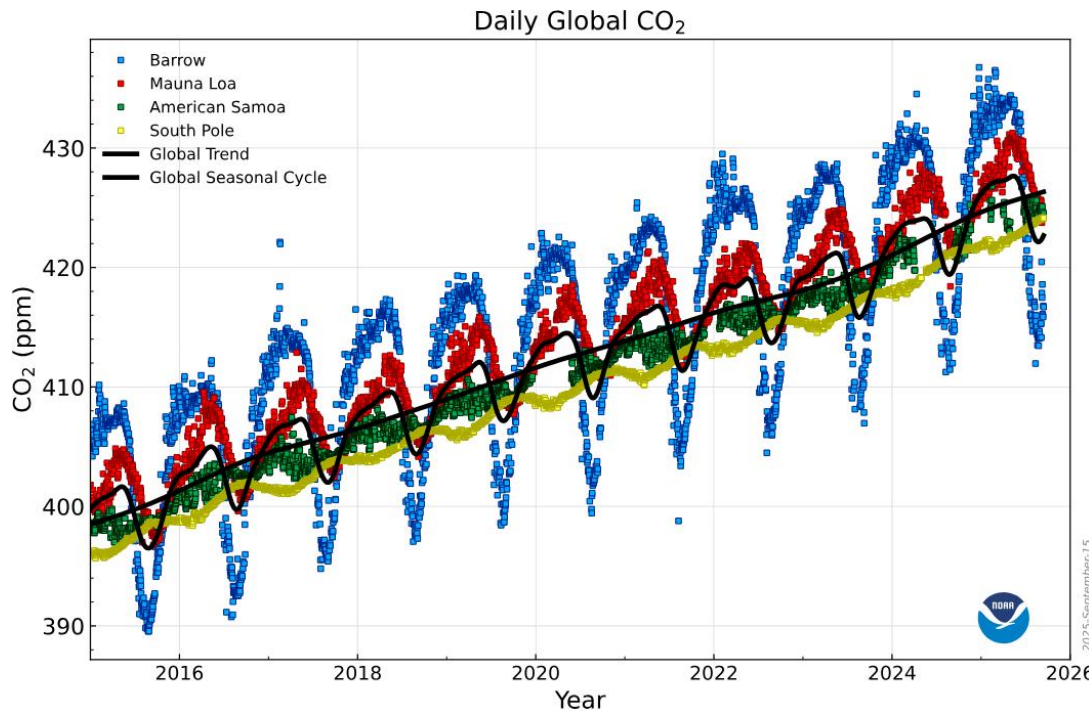
**The carbon cycle**



The **inventories** (bold font, units of Gt-C) and *fluxes* (italic font, units of Gt-C per year)

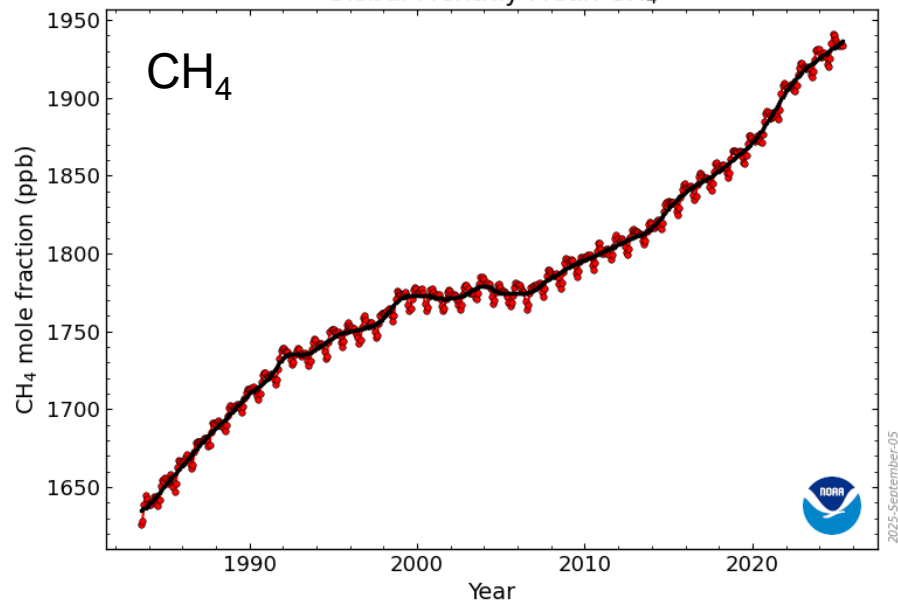
## 2. CO<sub>2</sub> Emission

**August 2025: 425.48 ppm**  
**August 2024: 422.99 ppm**  
*Last updated: Sep 05, 2025*

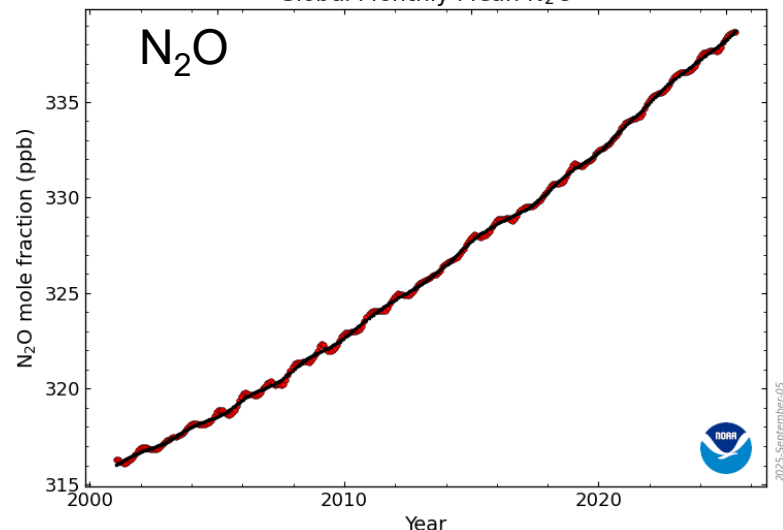


## 2. CO<sub>2</sub> Emission

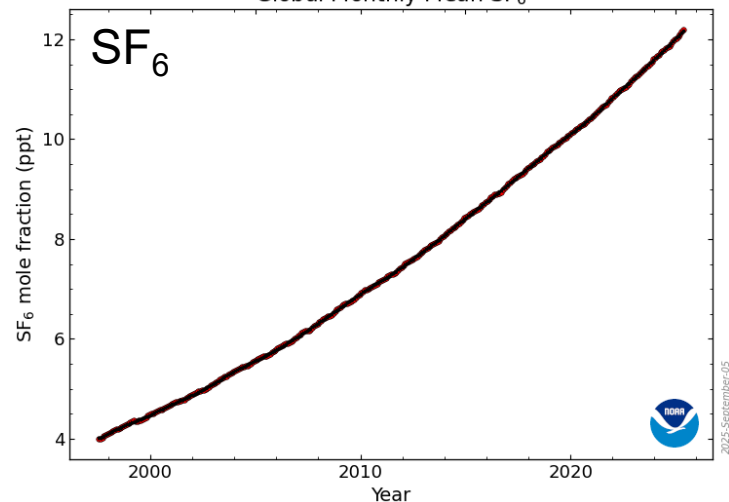
Global Monthly Mean CH<sub>4</sub>



Global Monthly Mean N<sub>2</sub>O



Global Monthly Mean SF<sub>6</sub>



### Global Warming Potential (GWP)

- CO<sub>2</sub>: 1 GWP (by definition)
- CH<sub>4</sub>: 27-30 GWP
- N<sub>2</sub>O: 273 GWP
- SF<sub>6</sub>: 23,500 GWP (high-GWP)

### 3. CO<sub>2</sub> - Climate interaction mechanisms

4  
400  
350  
300  
250  
200

**Jan 2021: 415 ppm**

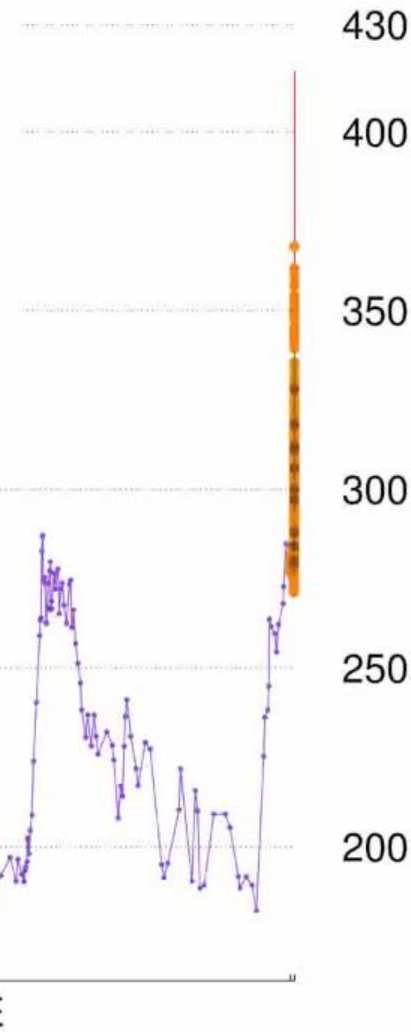
**Jan 1979: 336 ppm**

**Preindustrial: about 278 ppm**

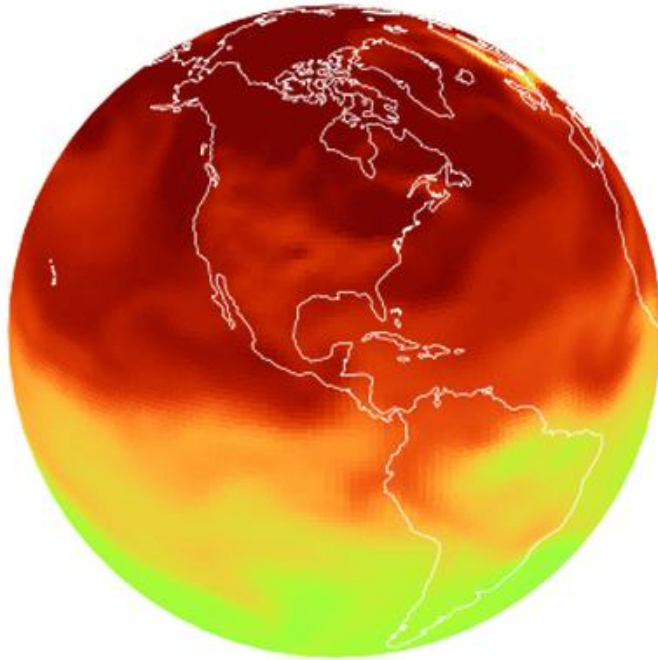
**Ice ages: about 185 ppm**

### Atmospheric CO<sub>2</sub> (ppm)

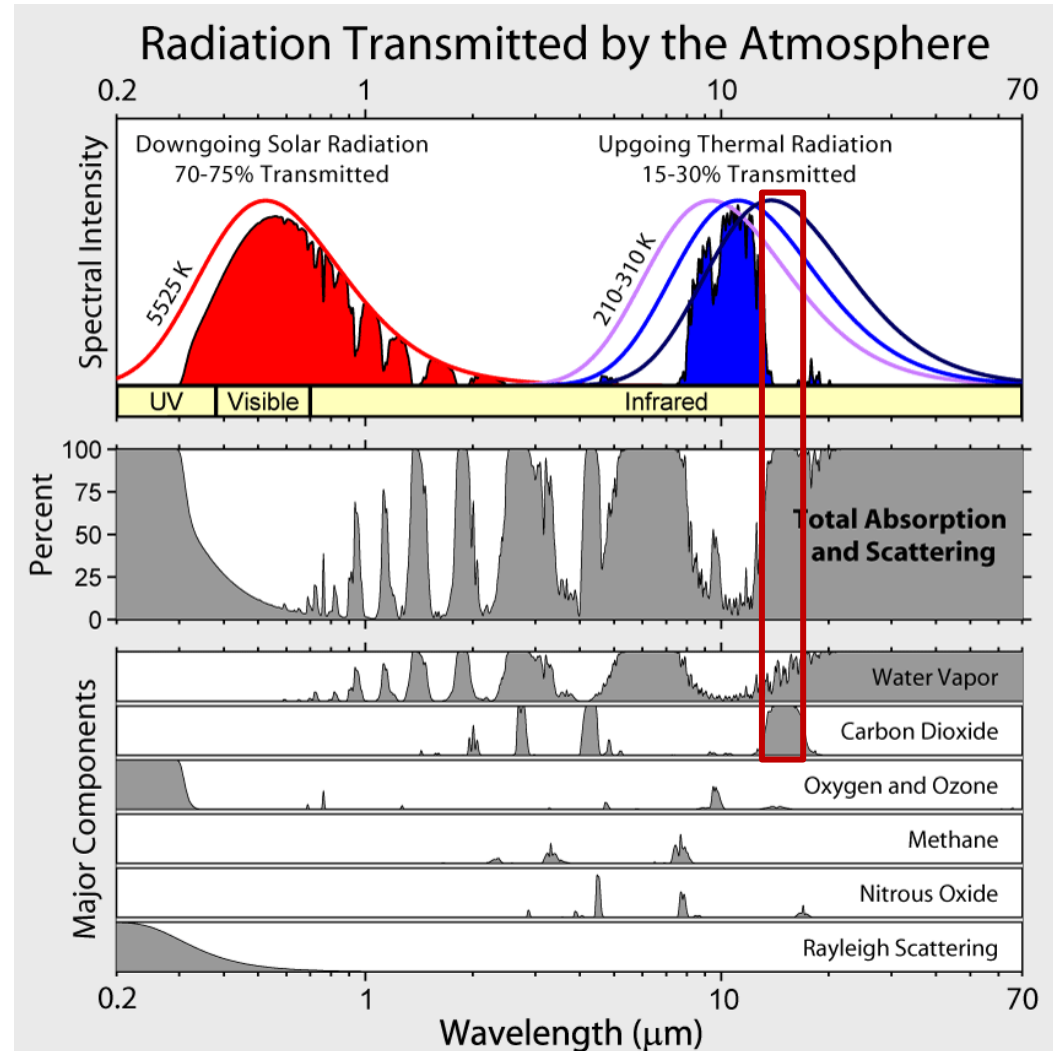
- GLOBALVIEW+CO<sub>2</sub> (1979–2021); <https://gml.noaa.gov/ccgg/obspack>
  - Keeling data (1958–1979): Scripps CO<sub>2</sub> program; <http://scrippsco2.ucsd.edu/>
  - Law dome ice core: Rubino et al., JGR 118 (2013), MacFarling Meure et al., GRL 33 (2006)
  - Siple ice core: Neftel et al., Nature 315 (1985)
  - Vostok ice core: Petit et al., Nature 399 (1999)
  - EPICA Dome C ice core: Siegenthaler et al., Science 310 (2005), Lüthi et al., Nature 453 (2008)
- yBCE = years before common era; kyBCE = thousands of years before common era  
Contact: andy.jacobson@noaa.gov



### 3. CO<sub>2</sub> - Climate interaction mechanisms

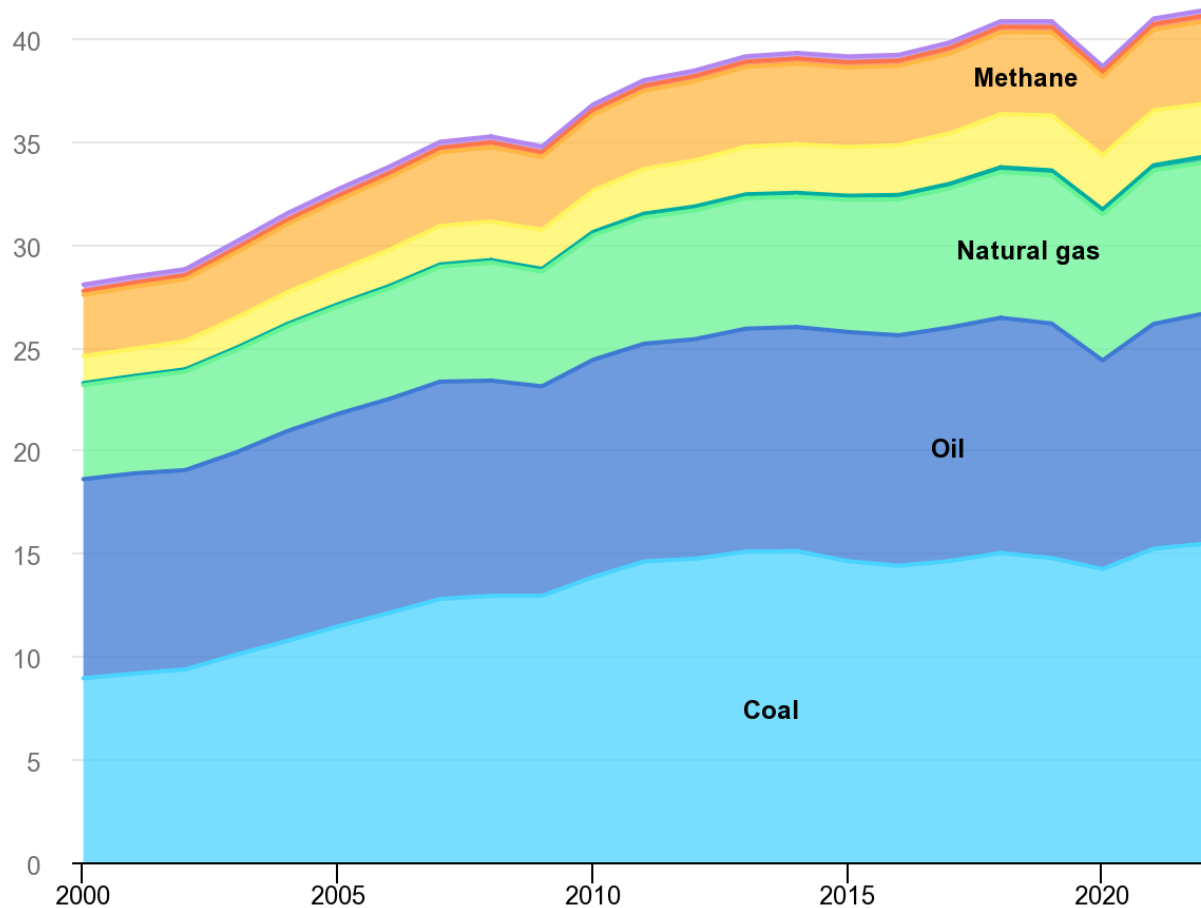


Effetto serra: cos'è.



**3. CO<sub>2</sub> - Climate  
interaction mechanisms**

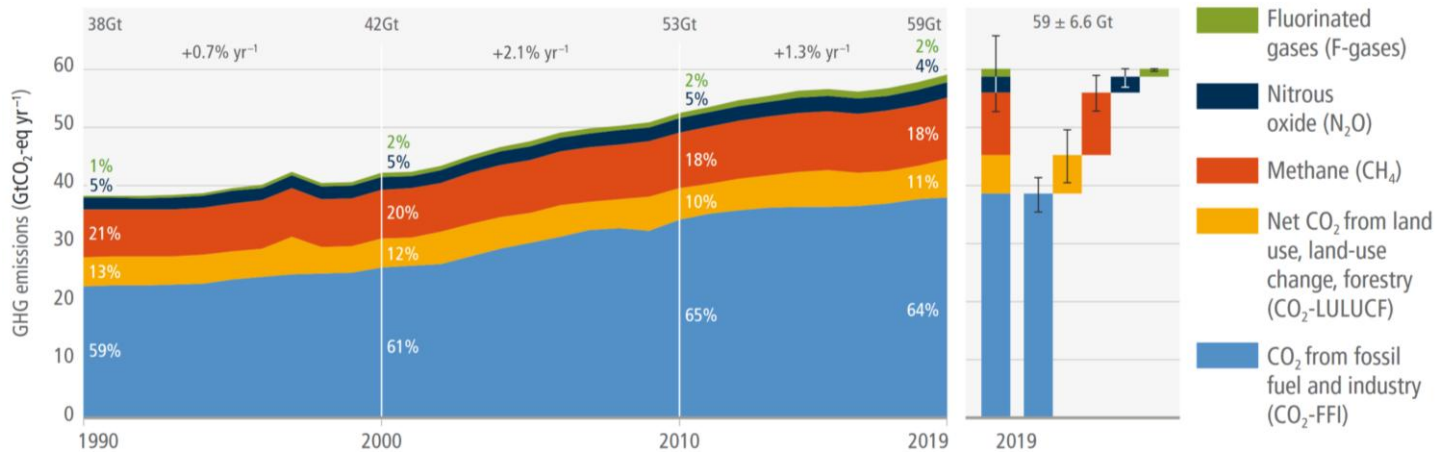
**Global energy-related greenhouse gas emissions**



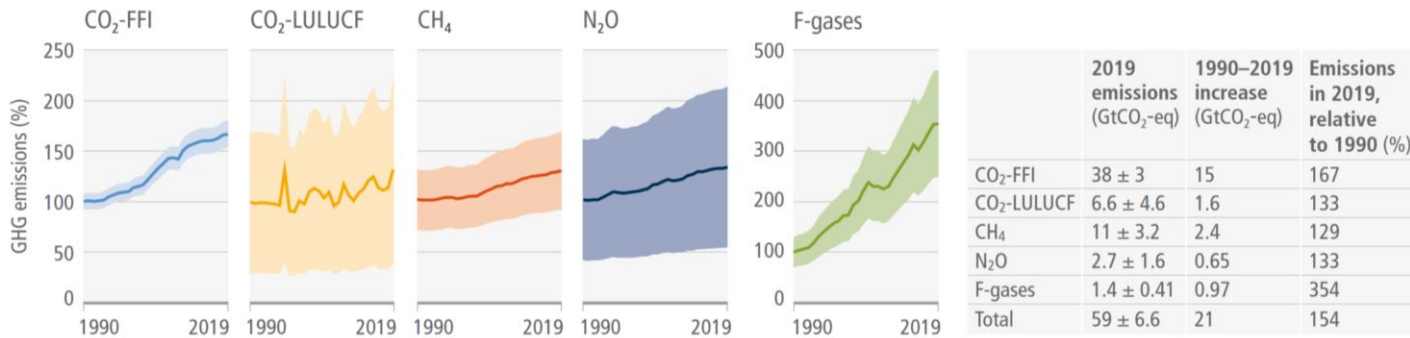
### 3. CO<sub>2</sub> - Climate interaction mechanisms

## Global energy-related greenhouse gas emissions

a. Global net anthropogenic GHG emissions 1990–2019 <sup>(5)</sup>



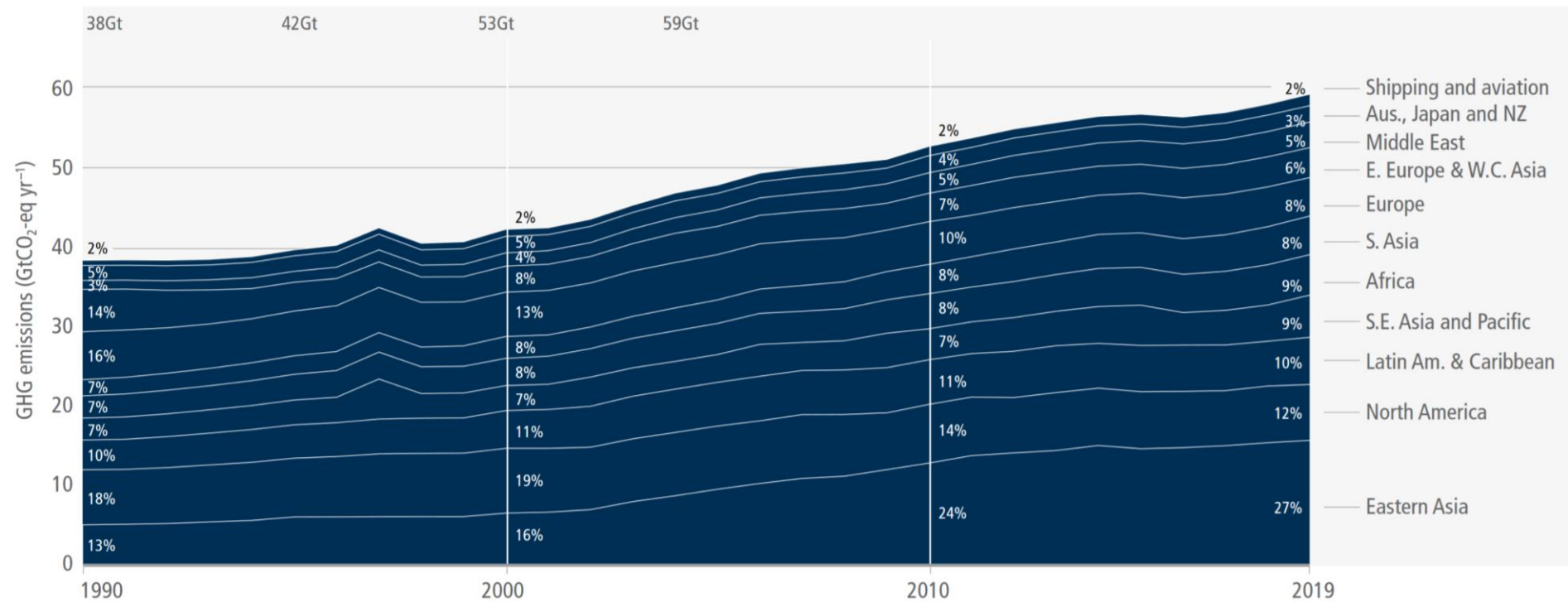
b. Global anthropogenic GHG emissions and uncertainties by gas – relative to 1990



3. CO<sub>2</sub> - Climate interaction mechanisms

# Global energy-related greenhouse gas emissions

(a) Global net anthropogenic GHG emissions by region (1990–2019)

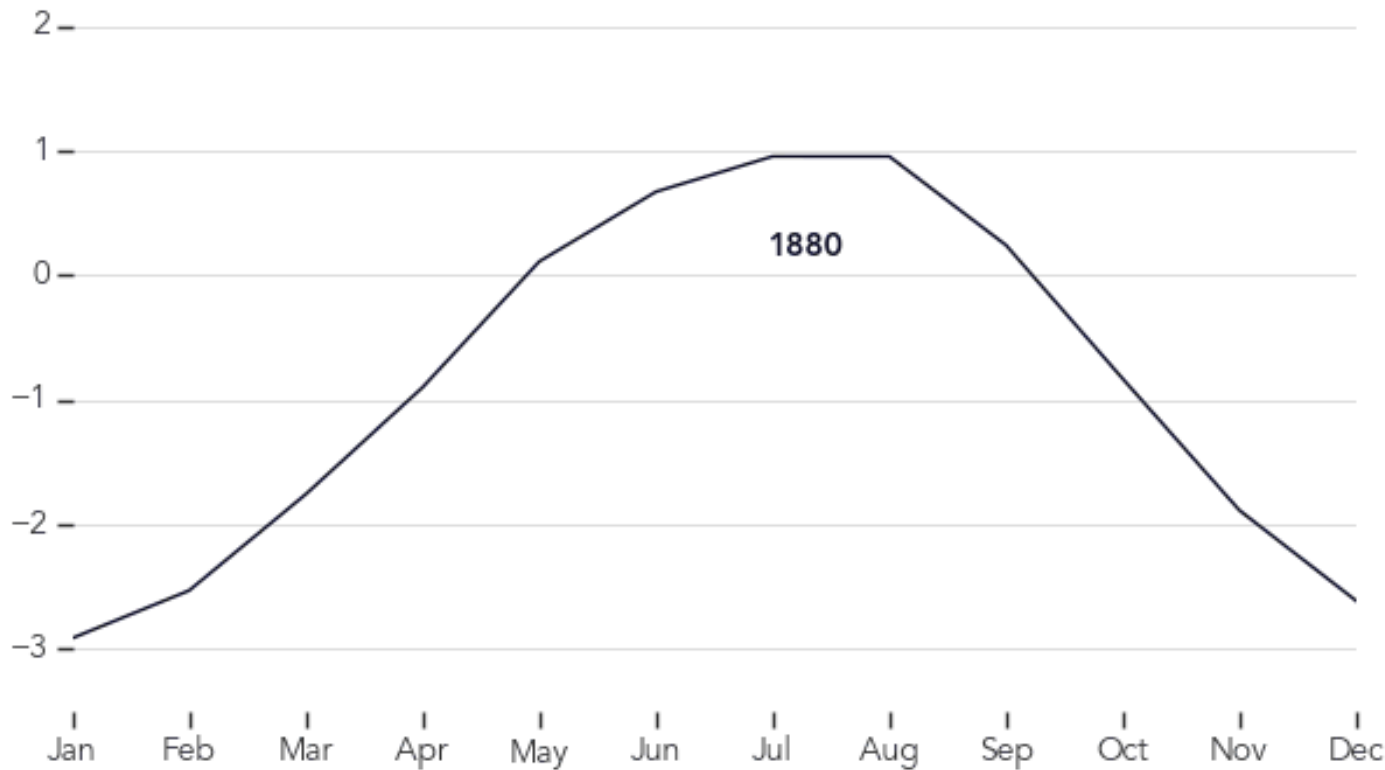


## 4. Climate change

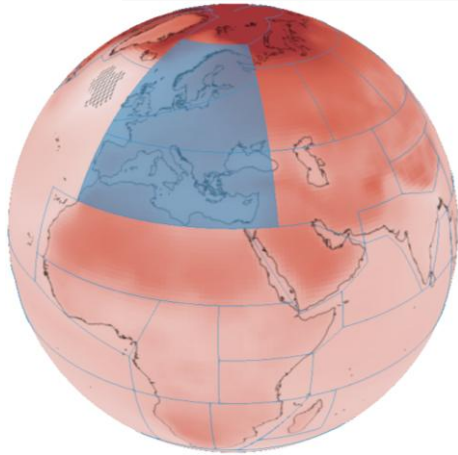
### The Present is Warmer than the Past

Difference from 1980-2015 annual mean, ( $^{\circ}\text{C}$ )

Record Years

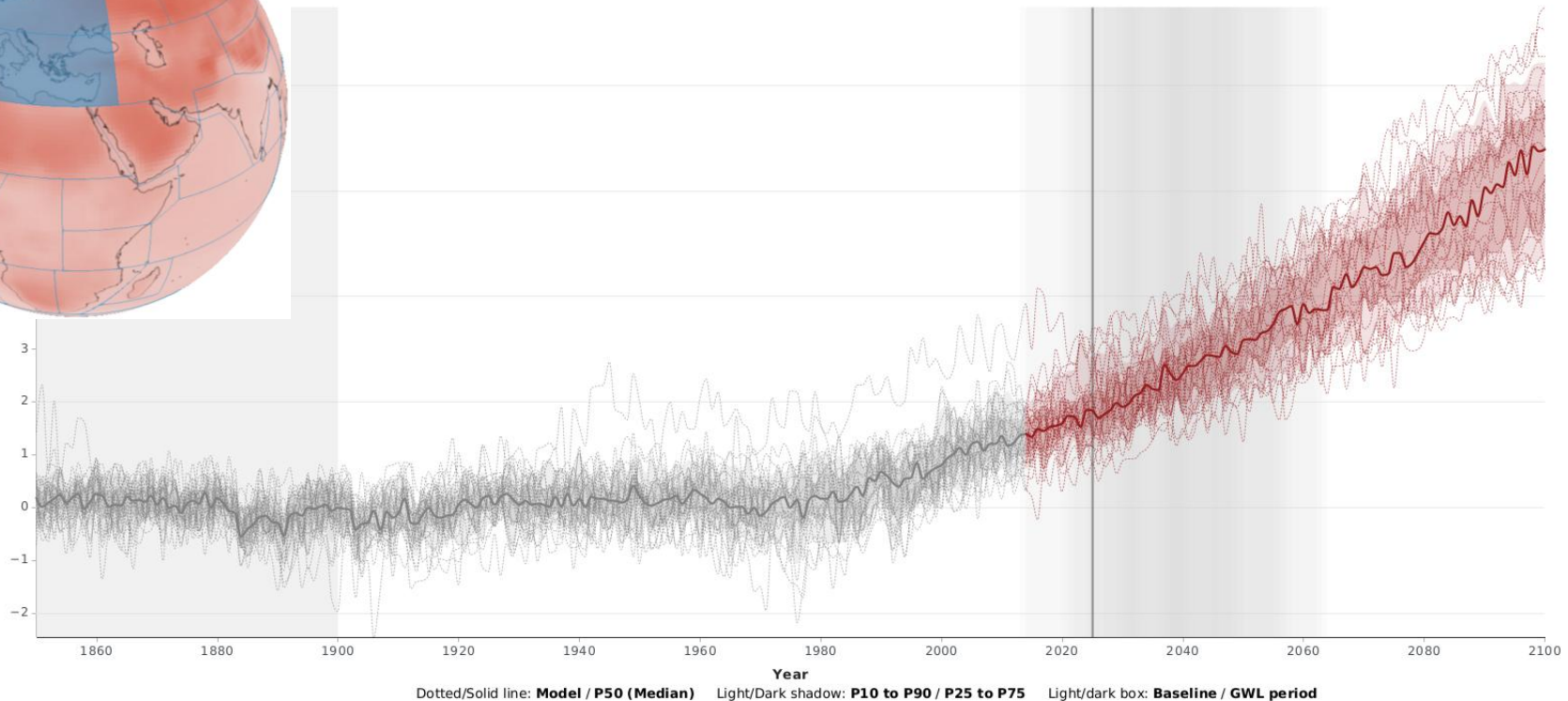


## 4. Climate change



**MEAN OF DAILY MEAN TEMPERATURE (°C) - CMIP6 - CHANGE - REL.  
TO 1850-1900 - WARMING 2°C - ANNUAL FOR N.EUROPE,  
WEST&CENTRAL-EUROPE AND MEDITERRANEAN**

Credit: C3S/ECMWF. Atlas version 2.1



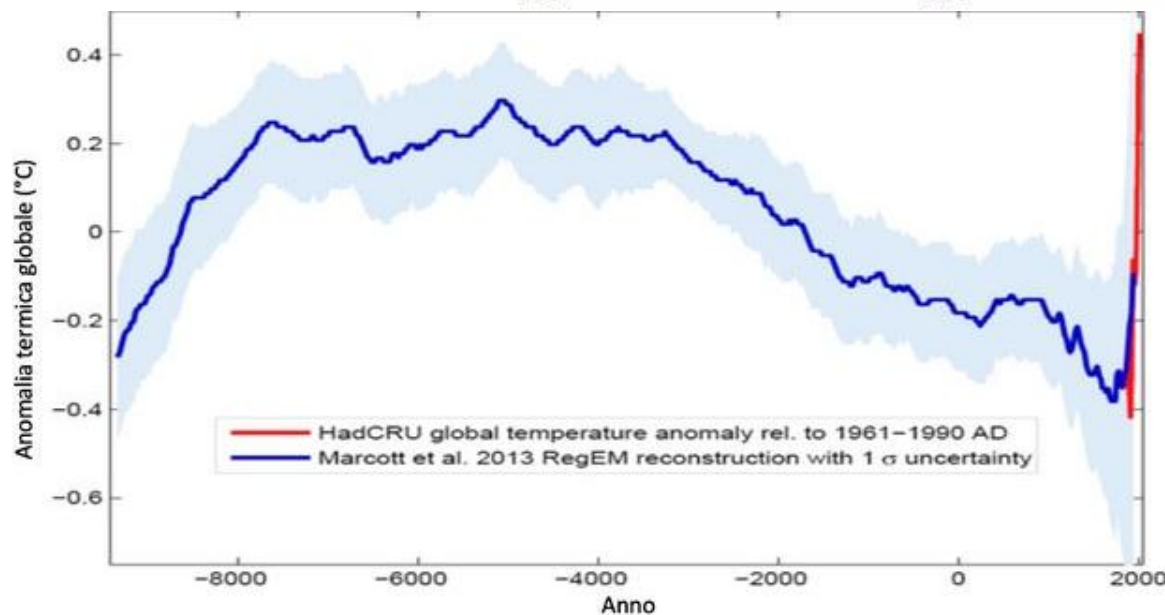
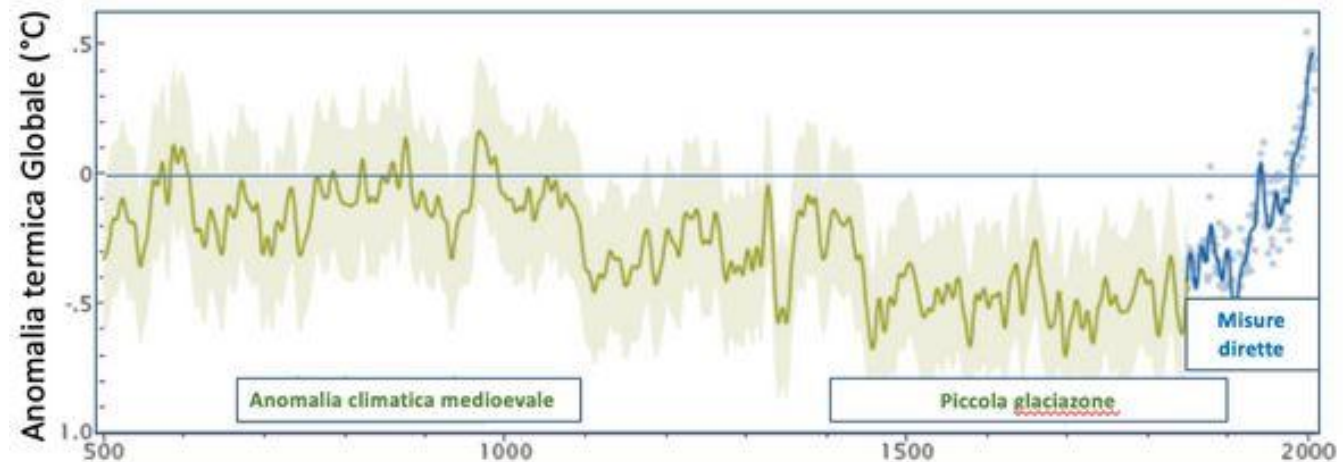
PROGRAMME OF  
THE EUROPEAN UNION



IMPLEMENTED BY

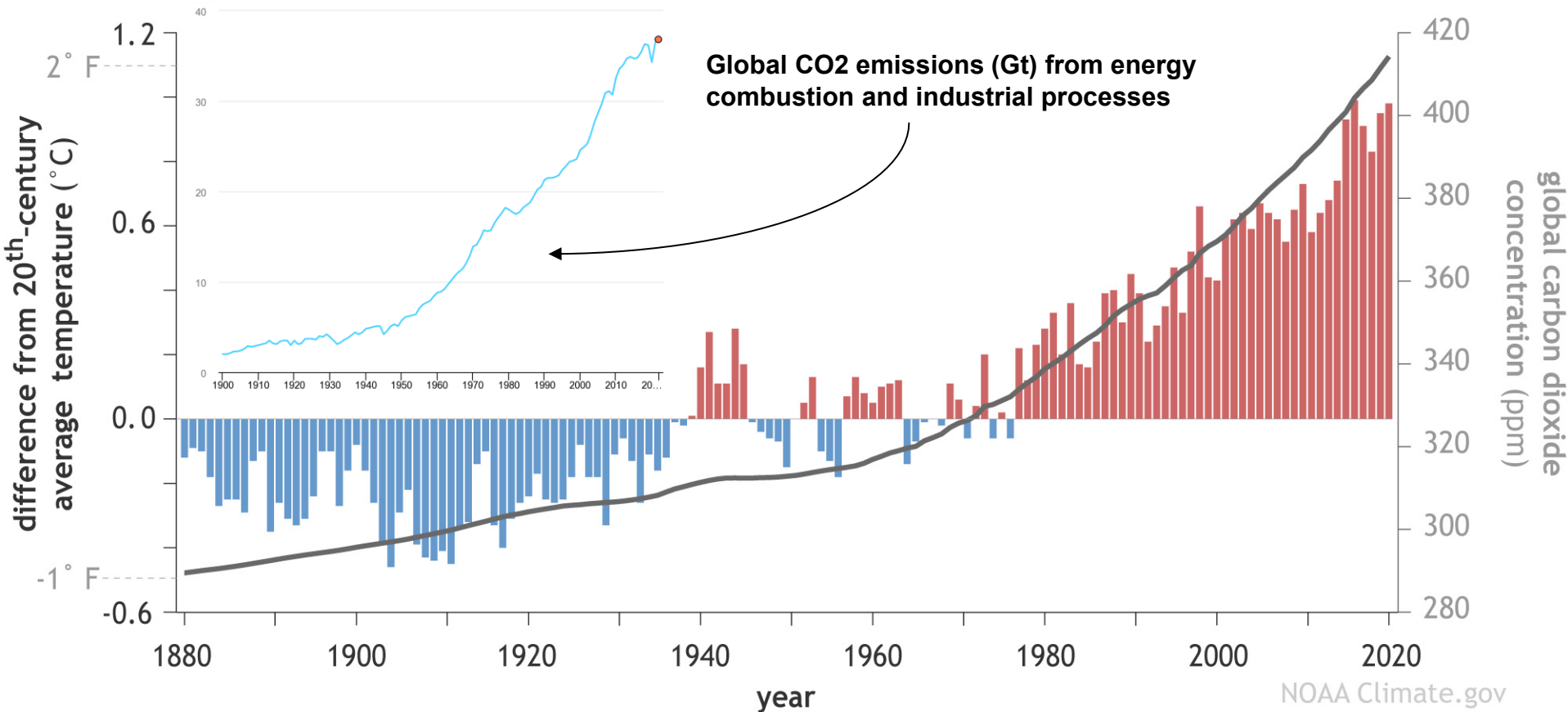


## 4. Climate change

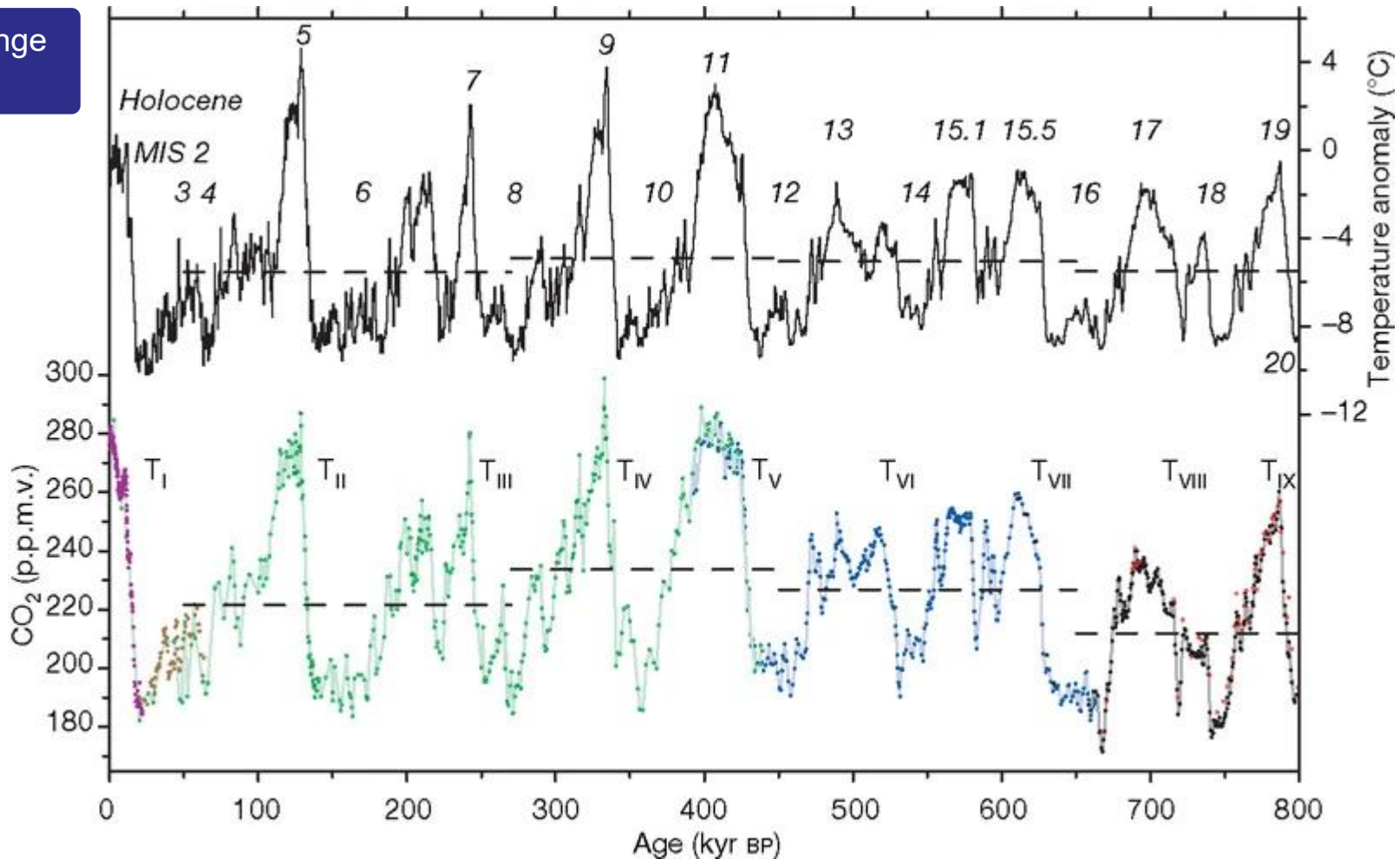


## 4. Climate change

### Global atmospheric carbon dioxide and surface temperature (1880-2020)



## 4. Climate change



Lüthi, D., Le Floch, M., Bereiter, B. *et al.* **High-resolution carbon dioxide concentration record 650,000–800,000 years before present.** *Nature* **453**, 379–382 (2008). <https://doi.org/10.1038/nature06949>

## 4. Climate change

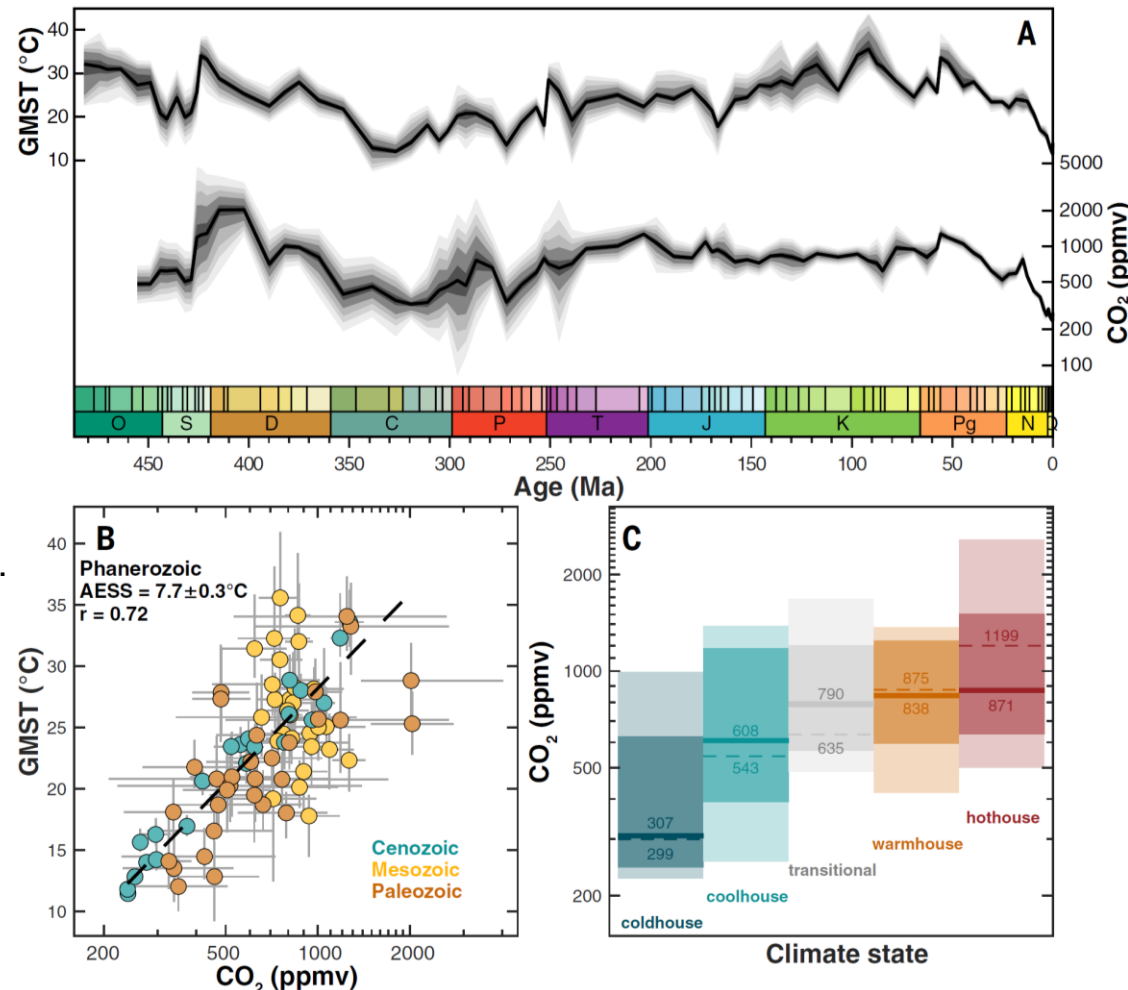
### The relationship between Phanerozoic temperature and atmospheric CO<sub>2</sub>.

(A) PhanDA GMST (top) and reconstructed atmospheric CO<sub>2</sub> (bottom).

Shading reflects percentiles,

(B) PhanDA GMST versus CO<sub>2</sub>, color-coded by geologic era.

(C) CO<sub>2</sub> ranges for each of the climate states. Light and dark bands show the 5th to 95th and 16th to 84th percentiles, respectively. The thick solid line shows the median value, and the dashed line shows the median, excluding data from the Mesozoic, where CO<sub>2</sub> is more uncertain.



Judd, E. J., Tierney, J. E., Lunt, D. J., Montañez, I. P., Huber, B. T., Wing, S. L., & Valdes, P. J. (2024). **A 485-million-year history of Earth's surface temperature.** *Science*, 385(6715), 0–8. <https://doi.org/10.1126/science.adk3705>

## 4. Climate change

### Atmospheric CO<sub>2</sub> exerts a dominant control on GMST, both today and in the geologic past

There is a significant positive correlation between CO<sub>2</sub> and GMST (somewhat surprising given that on Phanerozoic timescales).

The correlation between CO<sub>2</sub> levels and geologic eras is inconsistent across all periods (strong for the Cenozoic but weak for the Mesozoic era). This difference could be attributed to a lack of understanding of the CO<sub>2</sub> concentration (sometimes perplexingly) during these periods.

Non-CO<sub>2</sub> forcings factors playing a role in driving climate change:

- changes in solar luminosity
- other greenhouse gases (e.g.: higher methane production during Carboniferous and Permian could help offset the lower CO<sub>2</sub> and solar forcing during the LPIA, Late Paleozoic Ice Age)
- paleogeography (-> albedo\*)

\*From Ordovician, subaerially exposed continents constituted only ~15% of the total surface area of the planet (compared to ~30% today), with the value increasing quasi-linearly across the Paleozoic

Judd, E. J., Tierney, J. E., Lunt, D. J., Montañez, I. P., Huber, B. T., Wing, S. L., & Valdes, P. J. (2024). **A 485-million-year history of Earth's surface temperature.** *Science*, 385(6715), 0–8. <https://doi.org/10.1126/science.adk3705>

## Link utili

- [TERNA - Transparency Report](#)
- [Electricity maps in real time](#)
- [Trends in Atmospheric CO2 – NOAA](#)
- [Copernicus Interactive Climate Atlas](#)
- [Global Energy & CO2 Status Report 2019](#)
- [CO2 Emissions in 2023](#)

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Programmazione EU per la riduzione delle emissioni
- EU 20-20-20, Obiettivi 2030 e 2050
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